

Weighted Feature Graph with Adaptive Label Correlation for Partial Multi-Label Learning

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Abstract—Partial multi-label learning (PML) tackles the problem of learning from ambiguously annotated data, where instances are associated with candidate label sets containing both relevant and irrelevant tags. State-of-the-art methods typically employ feature graphs combined with label correlations to resolve this ambiguity through label distribution learning. However, current methods exhibit notable limitations: (1) they rely on unsupervised techniques to construct the feature graph, disregarding the influence of semantic information; (2) they derive label correlations directly from candidate label sets, ignoring the impact caused by noisy labels. To overcome these shortcomings, we propose a novel Partial Multi-Label Learning framework based on Weighted Feature Graph and Adaptive Label Correlation (PML-WFGALC). Specifically, a nonlinear predictive model enhanced by feature graph and label graph regularization is designed to recover the underlying label distribution of data. Meanwhile, the semantic similarity is integrated into the feature graph construction to better capture the intrinsic manifold structure of the data. In addition, an adaptive label graph learning mechanism is developed to facilitate classifier training through the dynamic exploration of label correlations. The framework jointly optimizes label distributions, label graph weights, and model parameters within a unified formulation, ensuring robust generalization capabilities. Experimental results on multiple benchmarks confirm the consistent superiority of PML-WFGALC compared to existing approaches.

Index Terms—Partial multi-label learning, weighted feature graph, label correlation, label distribution learning.

I. INTRODUCTION

MODERN machine learning applications frequently involve multi-label classification scenarios, where instances inherently belong to multiple categories simultaneously. This inherent complexity has positioned multi-label learning (MLL) [1] as a critical research direction, with successful implementations spanning diverse domains such as text mining and analysis [2], automated music categorization [3], and genomic function prediction [4]. However, traditional MLL methods rely on an idealized premise: each training instance is precisely annotated with all its relevant labels. In

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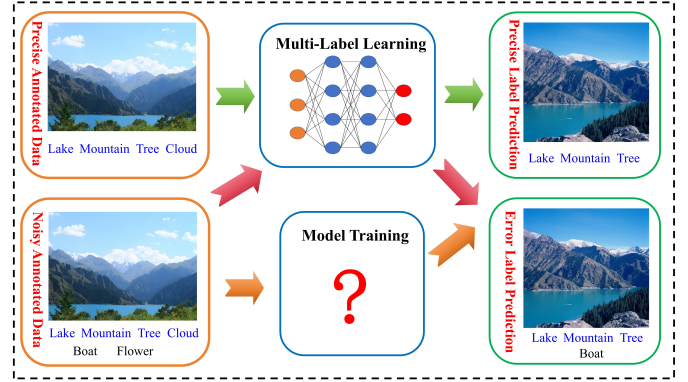


Fig. 1. The challenges posed by noisy labels to multi-label learning.

practical scenarios, this assumption is often violated—primarily due to the high costs of exhaustive annotation, the intrinsic difficulty of specific labeling tasks, and variations in annotator expertise. Consequently, each sample is typically assigned a candidate label set that contains both relevant and irrelevant labels, which may impair MLL model performance.

A common strategy for dealing with this problem is to treat all labels in the candidate set as relevant and apply existing MLL techniques to build a classifier. Nevertheless, such methods are vulnerable to being misled by noisy labels within the candidate set, as shown in Fig. 1. Moreover, label correlations deduced from noise contaminated candidate sets are often unreliable and inaccurate, resulting in suboptimal classification performance. To tackle these challenges, Xie et al. [5] introduced a method that assigns a confidence value to each label in the candidate set (estimating its likelihood of relevance to the corresponding instance) and formally defined this problem as Partial Multi-Label Learning (PML). As a form of weakly supervised learning, PML aims to develop multi-label classification models that can effectively handle noisy labels, improve classification accuracy, and enhance robustness—all without the need for precise manual annotations.

In recent years, a variety of PML methods have been developed, among which those based on label distribution learning (LDL) [6], [7], [8] have attracted widespread attention. Unlike traditional MLL, which uses binary logical labels to indicate the presence (1) or absence (0) of a label, LDL goes beyond binary relevance by modeling the degree of relevance of each label to the instance—represented as a continuous distribution

over all possible labels (e.g., probabilities summing to 1). This fine-grained representation allows LDL to capture nuanced relationships between instances and labels, which is particularly valuable in scenarios with ambiguous or noisy annotations. Therefore, LDL has shown promising results in PML. Methods based on LDL primarily recover the latent label distribution of data from candidate label sets by mining the data's topological structure and label correlations. Despite their promising outcomes, these methods still have several limitations that can adversely affect model performance. First, when exploring the topological structure of data, they only consider the similarity of samples in the feature space while neglecting their relationships in the label space. Second, mining label correlations directly from noise-contaminated candidate label sets often leads to inaccurate and unreliable results, as demonstrated in previous MLL research [9]. These issues underscore the need for more robust approaches to effectively handle noisy labels and incomplete annotations.

To address these challenges, we propose a novel method named Partial Multi-Label Learning with Weighted Feature Graph and Adaptive Label Correlation (PML-WFGALC). Specifically, PML-WFGALC builds a nonlinear model enhanced by feature-label dual graph regularization to estimate label distributions and identify high-confidence candidate labels. Meanwhile, PML-WFGALC integrates semantic similarity between samples into the k-nearest neighbors (KNN) process to construct the feature graph, ensuring that the feature graph reflects both the geometric structure and semantic relationships of the data, leading to a more accurate representation of the underlying data manifold. Subsequently, an adaptive learning mechanism is designed to dynamically update the label graph weights, ensuring the accuracy and reliability of label correlations. Finally, we propose a unified framework that jointly optimizes the label distribution, label graph weights, and model parameters via an efficient iterative strategy, significantly improving generalization performance. Our key contributions are listed as follows:

- A feature-label dual-graph-regularized nonlinear model is proposed, which jointly captures intrinsic feature-space structures and label correlations, enabling more precise label distribution estimation.
- A semantic similarity-based weighted KNN method is designed for feature graph construction, which outperforms conventional approaches by better preserving data geometry and semantic relationships.
- An adaptive label graph learning mechanism is designed to dynamically updates label correlations, which mitigates the limitations of static graphs derived from noisy candidate labels.
- Extensive experiments on 24 benchmark datasets (18 synthetic and 6 real-world) demonstrating that PML-WFGALC consistently surpasses SOTA methods in accuracy and robustness.

The remainder of this paper is structured as follows. Section II reviews related work in partial multi-label learning. Section III details the proposed PML-WFGALC method, including its optimization process. Section IV presents experimental results

and analyses. Finally, Section V concludes the paper.

II. RELATED WORK

With the advancement of big data technology, data resources have grown abundant, but the prevalence of noise has also increased-posing challenges to traditional learning models. Partial multi-label learning (PML), an emerging weakly supervised learning approach, addresses this issue by building classification models from samples with noisy labels. Current PML methods mainly fall into two categories: decoupled methods [10], [11] and embedded methods [12], [13].

Decoupled approaches divide label disambiguation and model training into two sequential stages: first, they employ strategies such as label propagation (e.g., PARTICLE [14]) or feature-manifold-generated label confidence matrices (e.g., DRAMA [15]) to filter noisy labels and obtain clean label information; second, they learn a prediction model on the disambiguated data. For example, Xu et al. [6] exploited the topological structure of the feature space and label correlations to recover label distributions, and then learned a predictive model by fitting a regularized multi-output regressor to the recovered label distributions. Yan et al. [16] employed a disambiguation network to identify noisy labels in the candidate label set, and then trained a multi-label prediction network to map features to their disambiguated labels. Similarly, Qian et al. [17] proposed a label enhancement method to first recover label confidence, and then constructed a noise-tolerant broad learning system for the prediction model. However, the performance of such methods highly relies on the accuracy of label confidence estimation in the first stage, as inaccurate confidence information can degrade the performance of the second-stage predictor.

In contrast, embedded approaches integrate label confidence calculation and predictor training into a unified framework for joint optimization. For example, Sun et al. [18] enhanced classifier performance by jointly leveraging global and local label information to enforce consistency between label coefficient matrices and predictive models; Hu et al. [19] proposed a decomposition framework that approximates candidate label matrices using compressed instance-classifier matrices while dynamically exploring latent label correlations to guide classifier training. Moreover, researchers have also explored feature-guided methods, which link noisy labels to feature information to enable simultaneous noise disambiguation and model learning. Xie et al. [20] formulated a parameter matrix as a composite of a multi-label classifier and a noise identifier, jointly optimizing both via a unified process that incorporates label correlation modeling and feature-aware noise estimation. Zhong et al. [21] constructed noise-guided discriminative regularization to handle the intra- and inter-class distances of noisy and true features, using noisy feature information to mine disambiguation label-specific features for true-value feature space construction. Methods such as those proposed by Sun et al. [22] and Yang [23] applied low-rank and sparse decomposition or l_1 -norm constraints to separate observed candidate label matrices into true label components and noise components.

With the rapid advancement of deep learning, several PML methods based on deep neural networks have also been proposed [24], [25], fully demonstrating their powerful capacity for feature representation and nonlinear modeling. These deep learning-driven approaches typically integrate label disambiguation into end-to-end network architectures, enabling adaptive extraction of discriminative features while suppressing noise. For example, Wang et al. [26] proposed a deep PML model named PLAIN, which uses instance-level and label-level similarities to recover label confidences and exploit label dependencies, and trains a deep model to fit the label confidences. Sun et al. [27] proposed a disambiguation method based on curriculum learning, which trains a deep neural network through consistency regularization to tackle partial multi-label image classification tasks. Li et al. [28] proposed a theory-inspired deep network method, which effectively addresses the problems of incomplete features and label noise in multi-view MLL by integrating feature extraction, cyclic consistency estimation, and label correlation regularization. These deep learning-driven PML methods excel in capturing complex patterns in high-dimensional data, making them particularly effective for large-scale or visually/audiorich datasets. However, they may suffer from overfitting to noisy labels without proper regularization, and their black-box nature can hinder interpretability of the label disambiguation process.

In this paper, we also propose an embedded method for PML. Different from existing PML methods, we uniquely incorporate semantic similarity into the k-nearest neighbors algorithm to construct a more accurate feature graph, which better captures both geometric and semantic data relationships. Additionally, we design an adaptive learning mechanism for the label graph weight matrix—enabling more accurate and reliable exploration of label correlations—instead of relying on fixed label graph construction methods. Overall, these differences endow PML-WFGALC with enhanced capabilities to handle noisy labels, and to improve the generalization performance of PML classification models.

III. PROPOSED METHOD

A. Notations

In this work, we define $\mathcal{X} \in \mathbb{R}^h$ as the h -dimensional feature space, and $\mathcal{L} = \{l_1, l_2, \dots, l_q\}$ as the finite labeling space comprising q distinct class labels. Consider a PML training dataset $\mathcal{P} = \{(\mathbf{x}_i, L_i) \mid 1 \leq i \leq N\}$, where each instance $\mathbf{x}_i \in \mathcal{X}$ is associated with a candidate label set $L_i \subseteq \mathcal{L}$. The fundamental PML assumption states that the true label set $\hat{L}_i$ for $\mathbf{x}_i$ satisfies $\hat{L}_i \subseteq L_i$, though $\hat{L}_i$ remains unknown during training. Additionally, we denote $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times h}$ as the feature matrix composed of all training samples, and represent logical labels via binary matrix $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N]^T \in \{0, 1\}^{N \times q}$, where $y_{ij} = 1$, if $l_j \in L_i$, otherwise $y_{ij} = 0$. For an arbitrary matrix $\mathbf{A}$, its (i, j) -th element is denoted as a_{ij} . Fig. 2 illustrates PML-WFGALC for feature graph and label correlation guided label disambiguation.

As depicted in Fig. 2, we convert the PML problem into a label distribution learning problem to resolve label ambiguity, and leverage the topological structure of the feature space and label correlations to direct the label distribution learning. Specifically, we incorporate the semantic similarity among samples into feature graph extraction. Subsequently, we introduce label graph reconstruction into the iteration of label distribution learning to identify noisy labels. Finally, we guide prediction modeling by extracting information from both the feature and label spaces.

B. Problem Formulation

1) *Label Distribution Learning*: In PML, label distribution pertains to the probability distribution over all possible labels for a given instance. Unlike traditional logical labels that merely signify the presence or absence of a label, label distributions capture the extent to which each label is relevant to the instance. Thus, retrieving the label distribution from the logical labels can effectively discriminate between relevant and irrelevant labels within the candidate label set.

Formally, for an instance $\mathbf{x}_i$, the description degree of label l_j is denoted as d_{ij} , which satisfies $d_{ij} \in [0, 1]$ and $\sum_{j=1}^q d_{ij} = 1$. These description degrees form a data structure $\mathbf{d}_i = [d_{i1}, d_{i2}, \dots, d_{iq}]$, which corresponds to the label distribution of sample $\mathbf{x}_i$. To recover the label distribution matrix $\mathbf{D}$ from the logical label matrix $\mathbf{Y}$, we establish a nonlinear mapping framework via the following parametric model:

$$\mathbf{d}_i = \varphi(\mathbf{x}_i) \mathbf{W} + \mathbf{b}^T, \quad (1)$$

where $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q]$ denotes the model weight matrix, and $\mathbf{b} \in \mathbb{R}^q$ represents the bias term. $\varphi(\mathbf{x}_i)$ refers a nonlinear mapping function that maps $\mathbf{x}_i$ to a higher dimensional Hilbert space. Subsequently, we devise the following objective function to optimize the predictive model:

$$\begin{aligned} \min_{\mathbf{D}, \mathbf{W}, \mathbf{b}} \quad & \|\Phi \mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}\|_F^2 + \lambda \|\mathbf{W}\|_F^2 \\ \text{s.t.} \quad & \mathbf{D} \mathbf{1}_q = \mathbf{1}_N, \mathbf{D} \geq \mathbf{0}, \end{aligned} \quad (2)$$

where $\Phi = [\varphi(\mathbf{x}_1), \varphi(\mathbf{x}_2), \dots, \varphi(\mathbf{x}_N)]^T$ denotes the transformed feature matrix; $\mathbf{1}_N$ and $\mathbf{1}_q$ respectively represent an N -dimensional and q -dimensional vector with all elements being 1; λ is the regularization coefficient.

2) *Feature Graph Guided Regularization*: For the training data $\mathbf{X}$, we construct a feature graph $G = \langle V, E \rangle$ over the training examples. Here, $V = \{\mathbf{x}_i \mid 1 \leq i \leq N\}$ corresponds to the set of vertices, and $E = \{(\mathbf{x}_i, \mathbf{x}_j) \mid \mathbf{x}_j \in \mathcal{N}_{\mathbf{x}_i}, j \neq i\}$ corresponds to the set of edges from $\mathbf{x}_i$ to $\mathbf{x}_j$, with $\mathcal{N}_{\mathbf{x}_i}$ denoting the k -nearest neighbors of $\mathbf{x}_i$. The majority of existing methods employ the traditional unsupervised KNN algorithm to determine $\mathcal{N}_{\mathbf{x}_i}$. This approach solely considers sample similarity in the feature space while ignoring the valuable guidance of semantic information. If the candidate label sets of two samples exhibit nearly no overlap (i.e., the semantic similarity between the two samples is extremely low), then these two samples ought not to have a neighboring relationship.

In light of this consideration, we put forward a weighted KNN method to incorporate semantic information into the

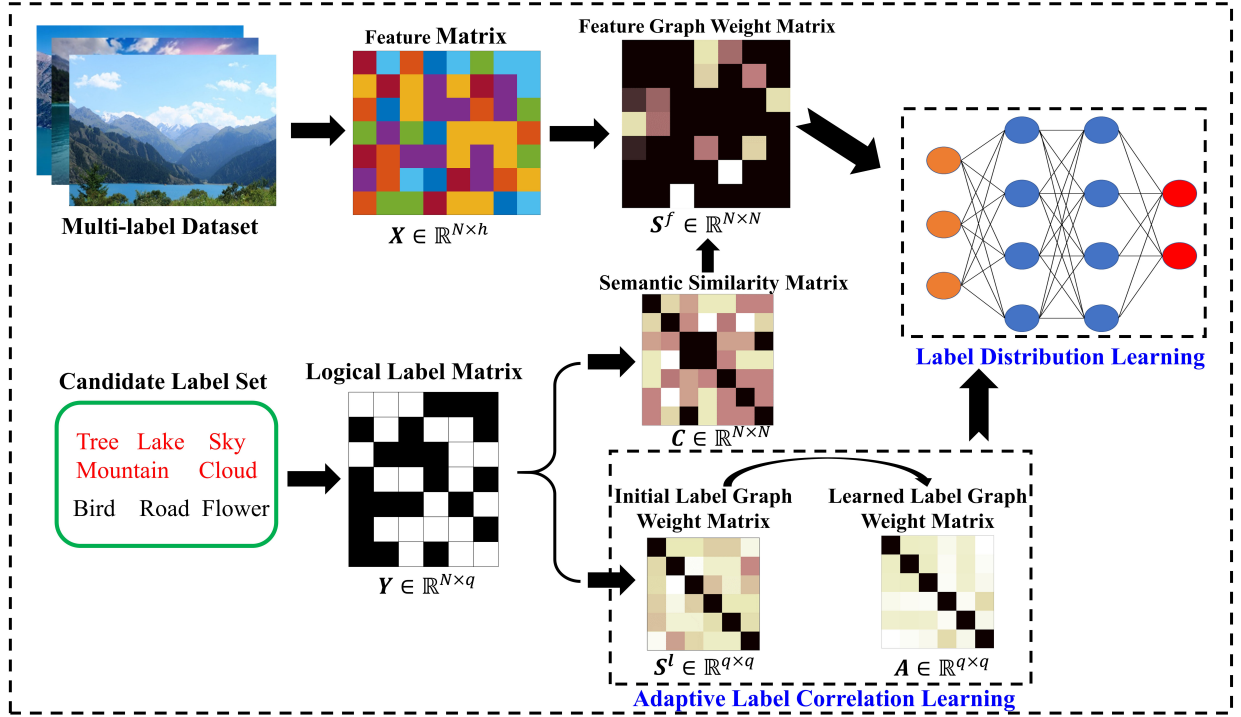


Fig. 2. The flow chart of the proposed PML-WFGALC.

k -nearest neighbor search. Specifically, we first compute the Euclidean distance dis_{ij} and the semantic similarity c_{ij} between any two samples x_i and x_j ($i \neq j$), where $c_{ij} = \frac{\mathbf{y}_i^T \mathbf{y}_j}{\|\mathbf{y}_i\|_2 \|\mathbf{y}_j\|_2}$ (i.e., the cosine similarity between candidate label vectors $\mathbf{y}_i$ and $\mathbf{y}_j$). Subsequently, the weighted distance between x_i and x_j is defined as $Wdis_{ij} = (1 - c_{ij})dis_{ij}$. For each training instance x_i , its neighborhood $\mathcal{N}_{x_i}$ consists of the k other training samples with the smallest weighted distances to it.

Subsequently, we solve the following linear least-square problem to acquire the feature graph weight matrix $\mathbf{S}^f \in \mathbb{R}^{N \times N}$:

$$\begin{aligned} & \sum_{i=1}^N \|\mathbf{x}_i - \sum_{\mathbf{x}_j \in \mathcal{N}_{x_i}} s_{ij}^f \mathbf{x}_j\|_2^2, \\ & s.t. \sum_{j=1}^N s_{ij}^f = 1, s_{ij}^f \geq 0 \quad (\forall \mathbf{x}_j \in \mathcal{N}_{x_i}), \\ & s_{ij}^f = 0, (\forall \mathbf{x}_j \notin \mathcal{N}_{x_i}), j = 1, 2, \dots, N, \end{aligned} \quad (3)$$

Finally, building upon the smoothness assumption which posits that the manifold structure present in feature space should be preserved in label space, we formulate the following manifold regularization term to guide label distribution learning:

$$\sum_{i=1}^N \|\mathbf{d}_i - \sum_{j=1, j \neq i}^N s_{ij}^f \mathbf{d}_j\|_2^2 = \text{trace}(\mathbf{D}^T \mathbf{M} \mathbf{D}), \quad (4)$$

where $\mathbf{M} = (\mathbf{I}_N - \mathbf{S}^f)^T (\mathbf{I}_N - \mathbf{S}^f)$, and $\mathbf{I}_N$ denotes an $N \times N$ identity matrix.

3) *Label Correlation Guided Regularization*: Mining label correlations is of utmost importance for constructing multi-label classifiers [29]. A prevalent approach is to construct a fixed label graph and compute the cosine similarity or label co-occurrence rate between labels to derive the label graph weight matrix. However, due to the presence of noisy labels in the candidate label sets, these methods are no longer dependable. Therefore, in this paper, we automatically update the label graph weight matrix during the label distribution learning process to acquire more robust and accurate label correlations. Additionally, given that label distributions should align with the underlying label correlations, we design the following manifold regularization term to guide the joint learning of label correlations and label distributions:

$$\begin{aligned} & \sum_{i=1}^q \sum_{j=1}^q a_{ij} \|\mathbf{d}^i - \mathbf{d}^j\|_2^2 + \|\mathbf{A} - \mathbf{S}^l\|_F^2 \\ & = \text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) + \|\mathbf{A} - \mathbf{S}^l\|_F^2 \\ & s.t. \mathbf{A} \geq \mathbf{0}, \mathbf{A}^T = \mathbf{A}, \text{diag}(\mathbf{A}) = \mathbf{0}, \mathbf{A} \mathbf{1}_q = \mathbf{1}_q, \end{aligned} \quad (5)$$

where $\mathbf{A} = \{a_{ij}\}_{i,j=1}^q$ is defined as the learned label graph weight matrix; $\mathbf{S}^l = \{s_{ij}^l\}_{i,j=1}^q$ denotes the initial label graph weight matrix, computed as $s_{ij}^l = \exp\left(-\frac{\|\mathbf{y}^i - \mathbf{y}^j\|_2^2}{\sigma^2}\right)$ —where σ is the bandwidth in Gaussian kernel function, and set to 1 in our experiments; $\mathbf{y}^i$ and $\mathbf{y}^j$ denote the i -th and j -th column vectors of the logical label matrix $\mathbf{Y}$ respectively; $\mathbf{L} = \hat{\mathbf{A}} - \mathbf{A}$ is the graph Laplacian matrix of $\mathbf{A}$, and $\hat{\mathbf{A}}$ is a diagonal matrix, whose diagonal elements are $\hat{a}_{ii} = \sum_{j=1}^q a_{ij}$.

In Eq. (5), the ideal matrix $\mathbf{D}$ is the representation of precise labels and contains label correlation information. Therefore, dynamically updating matrix $\mathbf{A}$ by leveraging matrix $\mathbf{D}$ through iterations enables $\mathbf{A}$ to carry more and more accurate

label correlations gradually. Through this label-level dynamic label graph weight matrix, label correlations are propagated to the label distribution $\mathbf{D}$, which in turn assists in training the classifier $\mathbf{W}$.

4) *Final Objective Function:* By integrating (2), (4), and (5) together, the final objective function of PML-WFGALC is as follows:

$$\begin{aligned} \min_{\mathbf{W}, \mathbf{b}, \mathbf{D}, \mathbf{A}} \quad & \|\Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2 + \lambda_2 \text{trace} \\ & (\mathbf{D}^T \mathbf{M} \mathbf{D}) + \lambda_3 (\text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) + \|\mathbf{A} - \mathbf{S}^l\|_F^2) \\ \text{s.t.} \quad & \mathbf{D} \mathbf{1}_q = \mathbf{1}_N, \mathbf{D} \geq \mathbf{0}, \mathbf{A} \geq \mathbf{0}, \mathbf{A}^T = \mathbf{A}, \\ & \mathbf{A} \mathbf{1}_q = \mathbf{1}_q, \text{diag}(\mathbf{A}) = \mathbf{0}, \end{aligned} \quad (6)$$

where λ_1 , λ_2 , and λ_3 represent the trade-off parameters.

C. Optimization

To solve Problem (6), we propose an iterative strategy to optimize $\{\mathbf{W}, \mathbf{b}\}$, $\mathbf{D}$ and $\mathbf{A}$ sequentially to obtain the optimal solution. First, we rewrite the objective function of PML-WFGALC as follows:

$$\begin{aligned} \min_{\mathbf{W}, \mathbf{b}, \mathbf{D}, \mathbf{A}} \quad & \|\Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2 \\ & + \lambda_2 \text{trace}(\mathbf{D}^T \mathbf{M} \mathbf{D}) + \lambda_3 (\text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) \\ & + \|\mathbf{A} - \mathbf{S}^l\|_F^2) + \frac{\mu}{2} \|\mathbf{D} \mathbf{1}_q - \mathbf{1}_N\|_2^2, \\ \text{s.t.} \quad & \mathbf{D} \geq \mathbf{0}, \mathbf{A} \geq \mathbf{0}, \mathbf{A}^T = \mathbf{A}, \mathbf{A} \mathbf{1}_q = \mathbf{1}_q, \text{diag}(\mathbf{A}) = \mathbf{0}, \end{aligned} \quad (7)$$

where $\mu > 0$ serves as a parameter for regulating the orthogonality condition. In practical applications, μ needs to be set large enough to ensure that the orthogonality is met.

1) *Update $\mathbf{W}$ and $\mathbf{b}$ with fixed $\mathbf{D}$ and $\mathbf{A}$:*

The sub-problem of (7) w.r.t. $\mathbf{W}$ and $\mathbf{b}$ is:

$$\min_{\mathbf{W}, \mathbf{b}} \|\Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2. \quad (8)$$

By introducing an error matrix $\mathbf{E} \in \mathbb{R}^{N \times q}$, we rewrite the above problem as:

$$\min_{\mathbf{W}, \mathbf{b}, \mathbf{E}} \|\mathbf{E}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2 \quad \text{s.t.} \quad \mathbf{E} = \Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}. \quad (9)$$

The Lagrangian function of Eq. (9) is:

$$\begin{aligned} \mathcal{L}(\mathbf{W}, \mathbf{b}, \mathbf{E}) = & \|\mathbf{E}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2 \\ & - \text{trace}(\mathbf{A}^T (\Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D} - \mathbf{E})) \end{aligned} \quad (10)$$

where $\mathbf{A} \in \mathbb{R}^{N \times q}$ represent the Lagrange multipliers. According to the KKT conditions, we have:

$$\begin{aligned} \frac{\partial \mathcal{L}(\mathbf{W}, \mathbf{b}, \mathbf{E})}{\partial \mathbf{W}} &= 2\lambda_1 \mathbf{W} - \Phi^T \mathbf{A} = \mathbf{0}, \\ \frac{\partial \mathcal{L}(\mathbf{W}, \mathbf{b}, \mathbf{E})}{\partial \mathbf{E}} &= 2\mathbf{E} + \mathbf{A} = \mathbf{0}, \\ \frac{\partial \mathcal{L}(\mathbf{W}, \mathbf{b}, \mathbf{E})}{\partial \mathbf{b}} &= \mathbf{A}^T \mathbf{1}_N = \mathbf{0}, \\ \frac{\partial \mathcal{L}(\mathbf{W}, \mathbf{b}, \mathbf{E})}{\partial \mathbf{A}} &= \Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D} - \mathbf{E} = \mathbf{0}. \end{aligned} \quad (11)$$

From these linear equations, we can obtain:

$$\mathbf{W} = \frac{\Phi^T \mathbf{A}}{2\lambda_1}, \mathbf{b}^T = \frac{\mathbf{1}_N^T \Gamma^{-1} \mathbf{D}}{\mathbf{1}_N^T \Gamma^{-1} \mathbf{1}_N}, \mathbf{A} = \Gamma^{-1} (\mathbf{D} - \mathbf{1}_N \mathbf{b}^T), \quad (12)$$

where $\Gamma = \left(\frac{1}{2\lambda_1} \mathbf{K} + \frac{1}{2} \mathbf{I}_{N \times N}\right)$, and $\mathbf{K} = \Phi\Phi^T$ is the kernel matrix with its element $k_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ according to the corresponding kernel function $\kappa(\cdot, \cdot)$. In this paper, we apply the Gaussian kernel function $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}{2\sigma^2}\right)$ to compute the kernel matrix $\mathbf{K}$, and the kernel bandwidth σ is set as the mean of the pairwise distance between pairs of observations. Then, by combining with the specified kernel function, the modeling output matrix can be expressed as $\mathbf{O} = \Phi\mathbf{W} + \mathbf{1}_N \mathbf{b}^T = \frac{1}{2\lambda_1} \mathbf{K} \mathbf{A} + \mathbf{1}_N \mathbf{b}^T$.

2) *Update $\mathbf{D}$ with fixed $\mathbf{W}$, $\mathbf{b}$ and $\mathbf{A}$:*

The sub-problem of (7) w.r.t. $\mathbf{D}$ is:

$$\begin{aligned} \min_{\mathbf{D}} \quad & \|\mathbf{O} - \mathbf{D}\|_F^2 + \lambda_2 \text{trace}(\mathbf{D}^T \mathbf{M} \mathbf{D}) \\ & + \lambda_3 \text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) + \frac{\mu}{2} \|\mathbf{D} \mathbf{1}_q - \mathbf{1}_N\|_2^2 \\ \text{s.t.} \quad & \mathbf{D} \geq \mathbf{0}. \end{aligned} \quad (13)$$

The Lagrangian function of this problem is

$$\begin{aligned} \mathcal{L}(\mathbf{D}) = & \|\mathbf{O} - \mathbf{D}\|_F^2 + \lambda_2 \text{trace}(\mathbf{D}^T \mathbf{M} \mathbf{D}) \\ & + \lambda_3 \text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) + \frac{\mu}{2} \|\mathbf{D} \mathbf{1}_q - \mathbf{1}_N\|_2^2 \\ & - \text{trace}(\Theta^T \mathbf{D}), \end{aligned} \quad (14)$$

where $\Theta = \{\theta_{ij}\} \in \mathbb{R}^{N \times q}$ is the Lagrangian multipliers for constraint $\mathbf{D} \geq \mathbf{0}$. By setting $\frac{\partial \mathcal{L}(\mathbf{D})}{\partial \mathbf{D}} = \mathbf{0}$ with the KKT condition $\theta_{ij} d_{ij} = 0$, we have:

$$[\mathbf{O} - \mathbf{D} + \lambda_2 \mathbf{M} \mathbf{D} + \lambda_3 \mathbf{D} \mathbf{L} + \mu/2 \mathbf{D} \mathbf{Q} - \mu/2 \mathbf{N}]_{ij} d_{ij} = 0, \quad (15)$$

where $\mathbf{Q}$ and $\mathbf{N}$ represent the all-ones matrices of size $q \times q$ and $N \times q$ respectively.

Since the matrices $\mathbf{O}$, $\mathbf{M}$ and $\mathbf{L}$ in Eq. (15) are mixed signed, to ensure that the solution $\mathbf{D}$ is non-negative, according to [30], we introduce $\mathbf{O} = \mathbf{O}^+ - \mathbf{O}^-$, $\mathbf{M} = \mathbf{M}^+ - \mathbf{M}^-$ and $\mathbf{L} = \mathbf{L}^+ - \mathbf{L}^-$, where $o_{ij}^+ = (|o_{ij}| + o_{ij})/2$ and $o_{ij}^- = (|o_{ij}| - o_{ij})/2$. Then, we obtain the updating rule for $\mathbf{D}$ as follows:

$$d_{ij} \leftarrow d_{ij} \sqrt{\frac{(\mathbf{O}^+ + \mathbf{D} + \lambda_2 \mathbf{M}^- \mathbf{D} + \lambda_3 \mathbf{D} \mathbf{L}^- + \mu/2 \mathbf{N})_{ij}}{(\mathbf{O}^+ + \lambda_2 \mathbf{M}^+ \mathbf{D} + \lambda_3 \mathbf{D} \mathbf{L}^+ + \mu/2 \mathbf{D} \mathbf{Q})_{ij}}}. \quad (16)$$

3) *Update $\mathbf{A}$ with fixed $\mathbf{W}$, $\mathbf{b}$ and $\mathbf{D}$:*

The sub-problem of (7) w.r.t. $\mathbf{A}$ is:

$$\begin{aligned} \min_{\mathbf{A}} \quad & \sum_{i=1}^q \sum_{j=1}^q a_{ij} \|\mathbf{d}^i - \mathbf{d}^j\|_2^2 + \|\mathbf{A} - \mathbf{S}^l\|_F^2 \\ \text{s.t.} \quad & \mathbf{A} \geq \mathbf{0}, \mathbf{A}^T = \mathbf{A}, \mathbf{A} \mathbf{1}_q = \mathbf{1}_q, \text{diag}(\mathbf{A}) = \mathbf{0}, \end{aligned} \quad (17)$$

which can be rewritten as:

$$\begin{aligned} \min_{\mathbf{A}} \quad & \|\mathbf{A} - \mathbf{T}\|_F^2 \\ \text{s.t.} \quad & \mathbf{A} \geq \mathbf{0}, \mathbf{A}^T = \mathbf{A}, \mathbf{A} \mathbf{1}_q = \mathbf{1}_q, \text{diag}(\mathbf{A}) = \mathbf{0}, \end{aligned} \quad (18)$$

where $\mathbf{T} = \mathbf{S}^l + \mathbf{D}^T \mathbf{D}$. This problem can obtain the optimal solution by solving the following two sub-problems [31]:

$$\min_{\mathbf{A}_1} \|\mathbf{A}_1 - \mathbf{T}\|_F^2, \text{ s.t. } \mathbf{A}_1 = \mathbf{A}_1^T, \mathbf{A}_1 \mathbf{1}_q = \mathbf{1}_q. \quad (19)$$

and

$$\min_{\mathbf{A}} \|\mathbf{A} - \mathbf{A}_1\|_F^2, \text{ s.t. } \mathbf{A} \geq \mathbf{0}, \text{diag}(\mathbf{A}) = \mathbf{0}. \quad (20)$$

According to [32], the optimal solution of Problem (19) is:

$$\mathbf{A}_1 = \mathbf{U} + \frac{q + \mathbf{1}_q^T \mathbf{U} \mathbf{1}_q}{q^2} \mathbf{1}_q \mathbf{1}_q^T - \frac{1}{q} \mathbf{U} \mathbf{1}_q \mathbf{1}_q^T - \frac{1}{q} \mathbf{1}_q \mathbf{1}_q^T \mathbf{U}, \quad (21)$$

where $\mathbf{U} = (\mathbf{T} + \mathbf{T}^T)/2$. And Problem (20) can be solved by

$$\mathbf{A} = \max(\mathbf{A}_1, 0), \text{diag}(\mathbf{A}) = \mathbf{0}. \quad (22)$$

Finally, we summarize the entire optimization algorithm for solving Problem (7) in Alg. 1. Compared with other LDL-based PML methods, PML-WFGALC incorporates semantic information into feature graph construction—thereby capturing a more accurate feature manifold structure. Furthermore, it mines more reliable label correlations by adaptively learning the label graph weight matrix, which reduces the interference of noisy labels on label correlations. Consequently, it can obtain more accurate label distributions and ideal classifiers.

Algorithm 1 PML-WFGALC

Input: Training feature matrix $\mathbf{X}$; Training candidate label matrix $\mathbf{Y}$; Number of nearest neighbors k ; Regularization parameters $\lambda_1, \lambda_2, \lambda_3$ and μ .

Output: Classifier matrix $\mathbf{W}$ and bias vector $\mathbf{b}$.

1. Initialize the label distribution matrix $\mathbf{D}$;
2. Compute the initial label graph weight matrix $\mathbf{S}^l$ and the graph Laplacian matrix $\mathbf{L}$;
3. Calculate the kernel matrix $\mathbf{K} = [\kappa(\mathbf{x}_i, \mathbf{x}_j)]_{N \times N}$;
4. Set the feature graph $G = \langle V, E \rangle$ based on the weighted distance;
5. Obtain the feature graph weight matrix $\mathbf{S}^f$ by solving Problem (3);

repeat

6. Update $\mathbf{A}$ and $\mathbf{b}$ based on (12);
7. Update model output $\mathbf{O} = \frac{1}{2\lambda_1} \mathbf{K} \mathbf{A} + \mathbf{1}_N \mathbf{b}^T$;
8. Update $\mathbf{D}$ according to (16);
9. Update $\mathbf{A}$ according to (21) and (22);

until Convergence or the maximum number of iterations is reached.

D. Theoretical Analysis

1) *Complexity Analysis:* In this subsection, we undertake a detailed analysis of the time complexity pertaining to Alg. 1. The overall computational cost of Alg. 1 is divided into two parts. Before entering the iteration, the algorithm mainly involves calculating the kernel matrix $\mathbf{K}$, the feature graph weight matrix $\mathbf{S}^f$, and the initial label graph weight matrix $\mathbf{S}^l$. The cost of computing $\mathbf{K}$ and $\mathbf{S}^l$ is $\mathcal{O}(hN^2)$ and $\mathcal{O}(q^2N)$ respectively. Calculating $\mathbf{S}^f$ requires solving a QP problem with N variables and constraints simultaneously, so the computational cost is $N\mathcal{O}_{QP}(N; N)$. After entering the iteration, the algorithm updates $\mathbf{A}$ and $\mathbf{b}$ at first, which involves matrix inversion operation, so the cost is $\mathcal{O}(N^3)$. Secondly, the cost for updating $\mathbf{D}$ is $\mathcal{O}(qN^2 + q^2N)$. Finally, the cost of updating $\mathbf{A}$ is $\mathcal{O}(q^2N)$. Since q is usually much smaller than h and N , the total cost of PML-WFGALC is $\mathcal{O}(hN^2 + q^2N + T_1(N^3)) + N\mathcal{O}_{QP}(N; N)$, where T_1 denotes the number of iterations.

2) *Convergence Analysis:* In this subsection, we prove the convergence of the proposed iterative procedure in Alg. 1 for optimizing Problem (7).

Proposition 1. *The updating rules listed in in Alg. 1 monotonically decrease the objective function value of (7) in each iteration.*

Proof: In the iterative optimization process, for $\{\mathbf{W}, \mathbf{b}\}$, $\mathbf{D}$ and $\mathbf{A}$, we update one while keeping the other two fixed. For ease of presentation, we define the following notation:

$$\begin{aligned} f(\mathbf{W}, \mathbf{b}, \mathbf{D}, \mathbf{A}) = & \|\Phi \mathbf{W} + \mathbf{1}_N \mathbf{b}^T - \mathbf{D}\|_F^2 + \lambda_1 \|\mathbf{W}\|_F^2 \\ & + \lambda_2 \text{trace}(\mathbf{D}^T \mathbf{M} \mathbf{D}) + \lambda_3 (\text{trace}(\mathbf{D} \mathbf{L} \mathbf{D}^T) \\ & + \|\mathbf{A} - \mathbf{S}^l\|_F^2) + \frac{\mu}{2} \|\mathbf{D} \mathbf{1}_q - \mathbf{1}_N\|_2^2. \end{aligned} \quad (23)$$

Let $(\mathbf{W}^t, \mathbf{b}^t, \mathbf{D}^t, \mathbf{A}^t)$ be the solution at the t -th iteration and $f(\mathbf{W}^t, \mathbf{b}^t, \mathbf{D}^t, \mathbf{A}^t)$ be the corresponding objective function value. At the $t+1$ -th iteration, when updating $\mathbf{W}$ and $\mathbf{b}$, $\mathbf{D}^t$ and $\mathbf{A}^t$ are fixed and the remaining objective function (i.e., Eq. (8)), is a standard convex optimization problem. It is easy to prove that $\mathbf{W}^{t+1}$ and $\mathbf{b}^{t+1}$ computed based on Eq. (12) are the optimal solutions to Problem (8). Therefore, we have:

$$f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^t, \mathbf{A}^t) \leq f(\mathbf{W}^t, \mathbf{b}^t, \mathbf{D}^t, \mathbf{A}^t). \quad (24)$$

Similarly, when updating $\mathbf{D}$, $\mathbf{W}^{t+1}$, $\mathbf{b}^{t+1}$ and $\mathbf{A}^t$ are fixed, and the remaining objective function (i.e., Eq. (13)) is also a standard convex optimization problem with constraints. Thus, $\mathbf{D}^{t+1}$ obtained by solving Problem (13) using the Lagrange multiplier method and KKT conditions is also the optimal solution. Then, we have:

$$f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^{t+1}, \mathbf{A}^t) \leq f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^t, \mathbf{A}^t). \quad (25)$$

Finally, when updating $\mathbf{A}$ with fixed $\mathbf{W}^{t+1}$, $\mathbf{b}^{t+1}$, and $\mathbf{D}^{t+1}$, it has been proved in [33] that iteratively solving Problems (19) and (20) can finally converge to the global optimal solution of the original Problem (18), namely, $\mathbf{A}^{t+1}$ computed by Eq. (21) and Eq. (22) is the optimal solution of Problem (18). Thus, we obtain

$$f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^{t+1}, \mathbf{A}^{t+1}) \leq f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^{t+1}, \mathbf{A}^t). \quad (26)$$

Based on Eq. (24), Eq. (25) and Eq. (26), we obtain

$$f(\mathbf{W}^{t+1}, \mathbf{b}^{t+1}, \mathbf{D}^{t+1}, \mathbf{A}^{t+1}) \leq f(\mathbf{W}^t, \mathbf{b}^t, \mathbf{D}^t, \mathbf{A}^t), \quad (27)$$

which reflects that the objective function value of Problem (7) will monotonically decreases using the updating rules in Alg. 1 and Proposition 1 is proved.

Per Proposition 1, the iterative approach outlined in Alg. 1 can converges to a stationary point. The proposed optimization algorithm is efficient. In the experimental section, we demonstrate that our algorithm achieves usually converges around a small number of iterations.

IV. EXPERIMENT

A. Experimental setup

1) *DataSets and Evaluation Metrics:* To comprehensively evaluate the proposed algorithm, we conduct experiments on six publicly available real-world PML datasets. To enhance

TABLE I
CHARACTERISTICS OF THE MLL DATASETS AND THE REAL-WORLD PML DATASETS.

Type	DataSet	Instance	Features	Labels	Avg.GLs	Avg.CLs
MLL Datasets	Bibtex	7395	1836	159	2.4019	-
	Birds	351	260	19	1.863	-
	Emotions	593	72	6	1.869	-
	Medical	978	1449	45	1.245	-
	Stackexchess	1675	585	227	2.411	-
	Yeast	2417	103	14	4.237	-
PML Datasets	Mirflickr	5000	100	7	1.771	3.353
	Music_emotion	6833	98	11	2.424	5.292
	Music_style	6839	98	10	1.440	6.043
	YeastBP	4257	6139	217	7.985	8.437
	YeastCC	2434	6139	50	3.399	3.436
	YeastMF	2594	6139	39	2.379	2.424

validation robustness, we further augment the evaluation by generating synthetic PML datasets—via controlled injection of random noisy labels into six MLL benchmark datasets. Specifically, the quantity of added noisy labels is set as $r\%$ of the number of ground-truth labels, with $r\%$ taking values from the set $\{50\%, 100\%, 150\%\}$. Table I provides detailed information about these datasets. In this table, Avg.GLs and Avg.CLs denote the average number of ground-truth labels and the average number of candidate labels per sample, respectively. Note that samples without ground-truth annotations in the Birds, YeastBP, YeastCC, and YeastMF datasets were excluded from our experiments to ensure evaluation validity.

We use five standard multi-label evaluation metrics to comprehensively assess our method’s performance. Among these metrics, there is one label-based metric, namely *Accuracy*, and four example-based metrics: *Avg Precision*, *Coverage*, *One Error*, and *Ranking Loss*. The detailed formal definitions of these metrics are provided in [34]. Regarding *Accuracy* and *Avg Precision*, a greater value indicates superior performance. In contrast, for the remaining three metrics, a lower value signifies better performance.

2) *Comparative Methods and Parameters Setting*: To demonstrate the effectiveness of our proposed PML-WFGALC, we compare it with nine benchmark methods: two partial multi-label feature selection (PMFS) methods—PMSNE (2025) [35] and LCFS-PML (2025) [36]—and seven state-of-the-art (SOTA) PML methods: NLR (2024) [23], PML-ND (2024) [21], LSNRLS (2024) [37], PLAIN (2023) [26], PML-NI (2022) [20], PAR-MAP (2021) [14], and PML-LRS (2019) [22]. For the comparative methods, we tune their hyperparameters within the ranges recommended in their respective references. In the case of our PML-WFGALC approach, for λ_1 , we explore values from the set $\{10^{-2}, 10^{-1}, \dots, 10^3\}$, while λ_2 and λ_3 are searched within the set $\{10^{-3}, 10^{-1}, \dots, 10^5\}$. The max number of iteration is set to 100. The number of nearest neighbors k is uniformly set to 20 across all methods. For the two PMFS methods, all features are selected. To tune the hyper-parameters, we identify the parameter values that maximize the *Avg Precision* of each comparative method on the testing data as the optimal settings. Additionally, we conduct five-fold cross-validation for all nine comparative methods on each dataset, recording the average results to ensure valid comparisons. All experiments were conducted using MATLAB R2019a on a machine with an Intel(R) Core(TM) i9-11900 @2.50 GHz CPU and 64 GB of RAM.

B. Performance Comparison

The experimental results on *Avg Precision*, *Coverage*, and *Ranking Loss* are listed in Tables II–IV, and the results on *Accuracy* and *One Error* are presented in the Appendix. In each table, the last row shows the average ranking of each method across all datasets; for each dataset, the method with the best result is ranked 1. To further assess the statistical significance of the performance differences, we conducted paired t-tests between our approach and the comparative methods on each dataset. Table V summarizes the number of cases where our method outperforms, shows no significant difference from, or is inferior to the comparative methods. Based on these results, we can draw the following observations:

- 1) Across all 120 evaluation scenarios (derived from 24 datasets and 5 metrics), PML-WFGALC attains the best results in 75.0% of the cases and ranks second in 23.3% of the cases. Consequently, PML-WFGALC secures a top-two ranking in 98.3% of the scenarios, strongly validating the effectiveness of the proposed method.
- 2) Compared with the other four metrics, PML-WFGALC achieves the best results in fewer cases for *Accuracy*. During the experiment, we observed that PML-WFGALC struggles to strike a good balance between *Avg Precision* and *Accuracy*: on many datasets, the better the *Avg Precision* result, the worse the *Accuracy* result. Although PML-WFGALC fails to yield the best performance on most datasets when compared with other baselines, it still achieves the highest avg_rank in terms of *Accuracy*.
- 3) As shown in Table V, across 1080 statistical comparisons involving nine competing algorithms and five evaluation metrics, PML-WFGALC outperforms its competitors in 73.8% of cases. In particular, across all metrics, PML-WFGALC wins against or ties with the baselines in over 96% of cases.
- 4) PMSNE and NLR serve as strong baselines. However, PML-WFGALC outperforms both by a significant margin, as the linear classifiers inherent to these baselines limit their overall effectiveness.
- 5) Compared to the deep learning method PLAIN, PML-WFGALC outperforms it significantly in 77.5% of cases and only loses to it in one case. The reason is that PLAIN only uses feature similarity to construct a feature graph and candidate label sets to construct a label graph, which severely limits its performance.
- 6) On the real-world PML datasets, PML-WFGALC attains the best results in 73.3% of the cases (derived from 6 datasets and 5 metrics). In particular, on the large-scale dataset (defined as a dataset with a large number of samples, high dimensionality, and a high number of labels) YeastBP, PML-WFGALC obtains the best results across all metrics and is superior to the best baselines by 12.6% and 13.9% in terms of *Coverage* and *Ranking Loss*, respectively. This demonstrates the effectiveness of PML-WFGALC in handling high volumes of data.

Furthermore, we conduct Friedman’s test [38] as a means of statistical analysis. This test is conducted to quantify

TABLE II
EXPERIMENTAL RESULTS COMPARING ALL APPROACHES IN TERMS OF *Avg Precision* (WITH BEST PERFORMANCE INDICATED IN BOLDFACE).

DataSets	Noise Ratio	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	50%	.5907±.0179	.5667±.0029	.5302±.0067	.5841±.0032	.5075±.0132	.5722±.0084	.4237±.0082	.5518±.0042	.4064±.0036	.5699±.0036
	100%	.5804±.0052	.5578±.0055	.5225±.0061	.5696±.0038	.4934±.0141	.5583±.0127	.4352±.0075	.5296±.0067	.4163±.0055	.5482±.0066
	150%	.5729±.0053	.5526±.0052	.5110±.0059	.5564±.0053	.4880±.0079	.5278±.0037	.4436±.0092	.5083±.0035	.4096±.0035	.5061±.0031
Birds	50%	.6358±.0342	.6134±.0467	.6037±.0306	.6319±.0429	.5916±.0219	.5928±.0279	.5799±.0289	.6261±.0337	.5454±.0275	.6038±.0402
	100%	.6190±.0275	.5898±.0368	.5953±.0473	.6117±.0374	.5492±.0156	.5516±.0326	.5704±.0299	.6030±.0286	.5524±.0255	.5845±.0336
	150%	.6210±.0264	.5912±.0306	.5892±.0455	.6036±.0365	.5592±.0270	.5461±.0187	.5728±.0216	.5991±.0259	.5953±.0243	.5691±.0457
Emotions	50%	.7956±.0266	.7230±.0119	.7543±.0283	.7758±.0251	.7658±.0183	.7505±.0181	.7847±.0316	.7665±.0233	.7670±.0371	.7595±.0176
	100%	.7880±.0259	.7406±.0169	.7594±.0137	.7622±.0106	.7644±.0066	.7565±.0140	.7636±.0221	.7442±.0290	.7499±.0157	.7618±.0099
	150%	.7267±.0153	.7111±.0287	.6935±.0292	.7112±.0798	.6947±.0106	.6855±.0234	.7232±.0365	.6664±.0296	.6913±.0207	.7140±.0265
Medical	50%	.8985±.0040	.8858±.0123	.7680±.0151	.8970±.0130	.8522±.0133	.8373±.0218	.8424±.0098	.8885±.0116	.8048±.0133	.8468±.0493
	100%	.8894±.0013	.8702±.0178	.7557±.0271	.8914±.0108	.8544±.0092	.7778±.0216	.8470±.0124	.8774±.0143	.8041±.0181	.8437±.0501
	150%	.8849±.0078	.8611±.0119	.7081±.0197	.8835±.0130	.8362±.0167	.7626±.0445	.8426±.0106	.8659±.0108	.7978±.0120	.8465±.0266
Stackexchess	50%	.4337±.0674	.3855±.0615	.3814±.0599	.4216±.0685	.4050±.0706	.3751±.0547	.2367±.0440	.3986±.0658	.2782±.0413	.3786±.0740
	100%	.4094±.0645	.3803±.0615	.3710±.0554	.4051±.0615	.3758±.0752	.3972±.0543	.2381±.0443	.3858±.0612	.2804±.0376	.3700±.0682
	150%	.4035±.0626	.3787±.0632	.3537±.0521	.3955±.0629	.3391±.0610	.3844±.0573	.2377±.0430	.3776±.0614	.2824±.0385	.3706±.0643
Yeast	50%	.7585±.0092	.7475±.0113	.5747±.0100	.6177±.0064	.6092±.0083	.7428±.0115	.7304±.0108	.7495±.0082	.7548±.0091	.5932±.0087
	100%	.7508±.0121	.7457±.0127	.5442±.0127	.6040±.0059	.5888±.0061	.7409±.0131	.7012±.0093	.7290±.0099	.7476±.0098	.5727±.0064
	150%	.7343±.0125	.7329±.0112	.5156±.0039	.5897±.0114	.5682±.0099	.7214±.0093	.6764±.0095	.6789±.0081	.7301±.0088	.5534±.0051
Mirflickr	/	.8434±.0065	.7887±.0083	.6852±.0085	.6917±.0077	.7347±.0040	.7282±.0356	.8044±.0107	.7897±.0062	.8576±.0068	.7586±.0089
Music_emotion	/	.6328±.0079	.5770±.0074	.5912±.0065	.6126±.0097	.5339±.0236	.5639±.0091	.6351±.0121	.6085±.0099	.5319±.0079	.5986±.0075
Music_style	/	.7369±.0051	.7047±.0030	.7233±.0050	.7322±.0039	.6661±.0015	.7152±.0054	.7191±.0027	.7362±.0043	.6580±.0041	.7301±.0047
YeastBP	/	.4372±.0832	.4264±.0929	.2230±.0557	.4077±.0908	.2402±.0429	.3412±.0719	.3426±.0691	.3430±.0745	.3044±.0603	.3631±.0880
YeastCC	/	.5974±.1018	.5825±.0927	.3219±.0412	.5638±.0912	.4490±.0807	.5043±.0784	.5277±.0687	.5030±.0705	.4839±.0706	.5246±.0780
YeastMF	/	.4749±.0476	.4701±.0493	.2756±.0298	.4512±.0458	.3699±.0364	.3886±.0267	.4412±.0420	.3936±.0350	.3375±.0171	.4231±.0446
avg_rank		1.17	4.67	7.71	3.33	7.25	6.54	6.17	4.75	7.38	6.08

TABLE III
EXPERIMENTAL RESULTS COMPARING ALL APPROACHES IN TERMS OF *Coverage* (WITH BEST PERFORMANCE INDICATED IN BOLDFACE).

DataSets	Noise Ratio	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	50%	.1408±.0031	.1370±.0035	.2164±.0042	.1700±.0033	.2459±.0174	.1935±.0129	.1930±.0060	.2216±.0023	.3437±.0066	.1954±.0027
	100%	.1554±.0039	.1510±.0014	.2242±.0053	.1886±.0036	.2712±.0189	.2058±.0144	.1873±.0047	.2433±.0047	.3012±.0074	.2200±.0040
	150%	.1687±.0041	.1607±.0024	.2341±.0086	.1993±.0062	.2887±.0049	.2375±.0064	.1827±.0055	.2574±.0056	.2980±.0080	.2583±.0049
Birds	50%	.2515±.0173	.2614±.0213	.2751±.0156	.2636±.0223	.2948±.0151	.3030±.0173	.3176±.0202	.2691±.0137	.3038±.0183	.2909±.0245
	100%	.2634±.0091	.2700±.0162	.2807±.0208	.2739±.0122	.3289±.0268	.3029±.0087	.3169±.0237	.2812±.0213	.2828±.0146	.2896±.0245
	150%	.2797±.0213	.2887±.0278	.2989±.0375	.2911±.0317	.3354±.0072	.3260±.0213	.3086±.0276	.2994±.0281	.2462±.0087	.3158±.0299
Emotions	50%	.3092±.0145	.3731±.0331	.3391±.0295	.3205±.0180	.3426±.0292	.3450±.0249	.3106±.0134	.3289±.0196	.3252±.0106	.3343±.0285
	100%	.3182±.0213	.3688±.0337	.3455±.0226	.3385±.0286	.3294±.0228	.3483±.0295	.3187±.0220	.3559±.0257	.3367±.0224	.3387±.0287
	150%	.3500±.0164	.3592±.0292	.3831±.0173	.3576±.0185	.4163±.0338	.3868±.0167	.3734±.0347	.4189±.0293	.3997±.0282	.3598±.0273
Medical	50%	.0340±.0073	.0404±.0069	.0764±.0088	.0348±.0073	.0552±.0108	.0612±.0087	.0499±.0076	.0390±.0065	.0738±.0077	.0720±.0513
	100%	.0372±.0080	.0483±.0071	.0826±.0119	.0376±.0094	.0527±.0059	.0821±.0070	.0483±.0058	.0460±.0064	.0795±.0048	.0641±.0402
	150%	.0397±.0062	.0532±.0096	.1050±.0072	.0425±.0076	.0614±.0096	.0773±.0073	.0568±.0092	.0505±.0091	.0807±.0033	.0602±.0204
Stackexchess	50%	.2436±.1263	.2450±.1410	.3481±.1179	.2936±.1058	.3142±.1425	.2924±.1193	.3686±.2126	.2965±.1137	.2492±.1728	.3417±.1042
	100%	.2487±.1308	.2577±.1394	.3508±.1045	.3080±.0995	.3562±.1637	.2695±.1327	.3741±.2104	.3115±.1080	.2505±.1660	.3632±.0993
	150%	.2577±.1347	.2632±.1377	.3534±.1098	.3132±.1026	.3912±.1387	.2861±.1350	.3841±.2118	.3219±.1137	.2521±.1495	.3555±.0979
Yeast	50%	.4603±.0079	.4635±.0085	.6508±.0112	.6633±.0105	.6822±.0075	.4672±.0131	.5005±.0091	.4857±.0063	.4732±.0087	.6438±.0068
	100%	.4732±.0058	.4759±.0070	.6635±.0096	.6811±.0087	.6854±.0102	.4744±.0081	.5342±.0040	.5169±.0068	.4982±.0055	.6520±.0067
	150%	.4727±.0098	.5041±.0082	.6710±.0074	.6717±.0065	.6726±.0061	.5101±.0192	.5647±.0081	.5605±.0098	.5126±.0040	.6842±.0085
Mirflickr	/	.2086±.0058	.2596±.0053	.2990±.0062	.2991±.0051	.3205±.0079	.2947±.0223	.2182±.0065	.2295±.0037	.2043±.0056	.2902±.0056
Music_emotion	/	.4039±.0075	.4331±.0072	.4197±.0061	.4099±.0054	.4985±.0081	.4281±.0056	.4028±.0095	.4085±.0054	.4865±.0074	.4152±.0061
Music_style	/	.1982±.0068	.2111±.0093	.2031±.0053	.2052±.0069	.2820±.0032	.2045±.0077	.2246±.0070	.1993±.0060	.2828±.0063	.2007±.0066
YeastBP	/	.3193±.0165	.3654±.0260	.6145±.0158	.3958±.0236	.5799±.0327	.4645±.0171	.3729±.0092	.4627±.0177	.5695±.0388	.3820±.0278
YeastCC	/	.2410±.0735	.2684±.0674	.4886±.0589	.2865±.0614	.3900±.0759	.3262±.0562	.2556±.0549	.3338±.0470	.3899±.0595	.2808±.0682
YeastMF	/	.2520±.0173	.2894±.0243	.4730±.0179	.3009±.0198	.3584±.0191	.3377±.0186	.2832±.0122	.3445±.0136	.3935±.0249	.3124±.0289
avg_rank		1.29	3.69	7.38	4.63	8.42	6.13	5.23	5.54	6.29	6.42

TABLE IV
EXPERIMENTAL RESULTS COMPARING ALL APPROACHES IN TERMS OF *Ranking Loss* (WITH BEST PERFORMANCE INDICATED IN BOLDFACE).

DataSets	Noise Ratio	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	50%	.0764±.0014	.0729±.0020	.1193±.0029	.0896±.0022	.1272±.0116	.1024±.0077	.1210±.0039	.1185±.0021	.1992±.0042	.1040±.0018
	100%	.0847±.0018	.0811±.0014	.1240±.0023	.1011±.0025	.1422±.0101	.1103±.0087	.1166±.0035	.1333±.0037	.1693±.0040	.1196±.0030
	150%	.0898±.0026	.0866±.0019	.1306±.0046	.1078±.0033	.1483±.0039	.1310±.0042	.1130±.0051	.1435±.0040	.1651±.0037	.1443±.0037
Birds	50%	.1568±.0110	.1659±.0198	.1761±.0162	.1643±.0194	.1895±.0215	.1967±.0122	.2133±.0196	.1689±.0154	.1998±.0186	.1842±.0241
	100%	.1662±.0098	.1747±.0151	.1830±.0197	.1733±.0124	.2179±.0120	.1985±.0088	.2108±.0172	.1809±.0183	.1902±.0143	.1875±.0240
	150%	.1778±.0209	.1862±.0251	.1949±.0331	.1883±.0266	.2253±.0124	.2225±.0135	.2082±.0224	.1962±.0222	.1549±.0140	.2121±.0260
Emotions	50%	.1737±.0179	.2426±.0171	.2093±.0238	.1883±.0140	.2046±.0153	.2108±.0124	.1798±.0210	.1964±.0160	.1939±.0208	.2047±.0126
	100%	.1843±.0182	.2312±.0147	.2078±.0139	.2047±.0072	.1976±.0059	.2130±.0162	.2095±.0186	.2262±.0247	.2102±.0086	.2042±.0072
	150%	.2373±.0120	.2433±.0170	.2705±.0180	.2410±.0174	.2898±.0147	.2741±.0110	.2522±.0363	.3048±.0197	.2847±.0198	.2472±.0154
Medical	50%	.0209±.0049	.0253±.0060	.0592±.0050	.0209±.0055	.0366±.0054	.0432±.0069	.0340±.0059	.0243±.0058	.0519±.0050	.0546±.0452
	100%	.0221±.0060	.0312±.0066	.0641±.0117	.0227±.0069	.0343±.0040	.0609±.0062	.0314±.0037	.0294±.0062	.0543±.0042	.0473±.0379
	150%	.0239±.0038	.0356±.0071	.0841±.0036	.0269±.0056	.0415±.0064	.0579±.0082	.0389±.0073	.0333±.0069	.0566±.0036	.0446±.0202
Stackexchess	50%	.1220±.0592	.1225±.0648	.1885±.0518	.1534±.0432	.1618±.0639	.1507±.0516	.2017±.1045	.1540±.0529	.1324±.0822	.1844±.0447
	100%	.1252±.0595	.1297±.0638	.1941±.0515	.1643±.0398	.1881±.0762	.1354±.0577	.2037±.1030	.1658±.0473	.1342±.0794	.1992±.0382
	150%	.1332±.0628	.1345±.0629	.1972±.0453	.1698±.0423	.2126±.0661	.1467±.0624	.2085±.1036	.1725±.0493	.1365±.0710	.1961±.0392
Yeast	50%	.1736±.0088	.1772±.0097	.3724±.0102	.3554±.0041	.3529±.0049	.1796±.0100	.2048±.0093	.1836±.0074	.1763±.0068	.3690±.0064
	100%	.1797±.0088	.1816±.0096	.3883±.0089	.3617±.0042	.3623±.0046	.1845±.0107	.2327±.0074	.2040±.0075	.1847±.0066	.3794±.0045
	150%	.1927±.0084	.2002±.0091	.4011±.0052	.3638±.0079	.3750±.0063	.2079±.0118	.2554±.0054	.2459±.0044	.1965±.0064	.4055±.0094
Mirflickr	/	.0945±.0046	.1796±.0068	.2224±.0082	.2446±.0042	.2313±.0077	.2724±.0259	.1190±.0076	.1263±.0083	.0914±.0066	.2011±.0080
Music_emotion	/	.2336±.0067	.2739±.0061	.2588±.0061	.2442±.0068	.3192±.0120	.2742±.0088	.2377±.0081	.2449±.0067	.3199±.0059	.2534±.0059
Music_style	/	.1387±.0038	.1533±.0057	.1446±.0035	.1446±.0046	.2216±.0018	.1465±.0047	.1641±.0036	.1393±.0036	.2228±.0040	.1417±.0041
YeastBP	/	.1577±.0330	.1832±.0444	.3091±.0389	.2011±.0455	.2837±.0143	.2452±.0443	.2087±.0328	.2458±.0455	.2528±.0184	.2107±.0431
YeastCC	/	.1476±.0682	.1672±.0622	.3040±.0402	.1804±.0574	.2370±.0614	.2138±.0506	.1676±.0430	.2198±.0483	.2309±.0495	.1875±.0661
YeastMF	/	.1840±.0170	.2185±.0310	.3513±.0261	.2294±.0238	.2583±.0221	.2641±.0116	.2143±.0151	.2713±.0165	.2833±.0126	.2473±.0361
avg_rank		1.23	3.67	7.69	4.04	7.79	6.29	5.83	5.54	6.38	6.54

TABLE V
WIN/TIE/LOSS COUNTS OF PAIRWISE T-TEST (AT 0.05 SIGNIFICANCE LEVEL) ON PML-WFGALC AGAINST EACH COMPARING APPROACH.

	Ours against									In Total
	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS	
<i>Accuracy</i>	7/15/2	22/2/0	13/9/2	23/1/0	12/10/2	20/3/1	21/3/0	24/0/0	14/10/0	156/53/7
<i>Avg precision</i>	15/9/0	20/4/0	15/9/0	22/2/0	22/2/0	21/3/0	19/5/0	20/3/1	17/7/0	171/44/1
<i>Coverage</i>	14/10/0	21/3/0	16/8/0	23/1/0	22/2/0	19/5/0	19/5/0	17/6/1	19/5/0	170/45/1
<i>One error</i>	12/11/1	18/6/0	8/16/0	16/8/0	21/3/0	14/10/0	15/9/0	15/8/1	15/9/0	134/80/2
<i>Ranking loss</i>	12/10/2	20/4/0	15/9/0	22/2/0	23/1/0	20/4/0	20/4/0	16/8/0	18/6/0	166/48/2
In Total	60/55/5	101/19/0	67/51/2	105/15/0	100/18/2	93/26/1	94/26/0	92/25/3	83/37/0	797/270/13

TABLE VI
SUMMARY OF THE FRIEDMAN STATISTICS FOR EACH EVALUATION METRIC AND THE CRITICAL VALUE AT A SIGNIFICANCE LEVEL $\alpha = 0.05$ FOR PML-WFGALC, WHERE THE NUMBER OF COMPARATIVE BASELINES IS 10 AND THE NUMBER OF DATASETS IS 24.

Metrics	F_F	Critical Value($\alpha = 0.05$)
<i>Accuracy</i>	38.187	1.925
<i>Avg Precision</i>	20.140	
<i>Coverage</i>	17.789	
<i>One Error</i>	15.269	
<i>Ranking Loss</i>	18.111	

performance differences between our proposed approach and the comparative baselines, thereby validating the superiority of PML-WFGALC. As shown in Table VI, for each evaluation metric, the F_F statistics are substantially larger than their corresponding critical values. As a result, the null hypothesis, which assumes homogeneous performance across all the comparative methods, can be firmly rejected at the 0.05 significance level. This rejection strongly substantiates the existence of significant performance disparities among the evaluated approaches.

Consequently, we conduct the post-hoc Nemenyi test [38] to perform a more in-depth analysis of the performance differences between our approach and the comparative algorithms. In this test, PML-WFGALC is set as the control algorithm, and the critical difference (CD) is used to measure the disparity in average ranks between PML-WFGALC and the other comparative baselines. Specifically, if the absolute difference in average ranks between PML-WFGALC and a comparative method exceeds one CD (in this study, with 10 comparative methods and 24 datasets, $CD = 2.7651$), there is a statistically significant performance difference between PML-WFGALC and the corresponding baseline.

Fig. 3 presents the CD diagrams for each evaluation metric. In these diagrams, the horizontal axis represents the average ranking of each method, arranged from left to right in ascending order (i.e., lower ranks—indicating better performance—are positioned on the right). In each subfigure, comparative baselines whose average rank difference with PML-WFGALC falls within a single CD interval are connected to PML-WFGALC by a red line, indicating that their performance differences relative to PML-WFGALC are statistically negligible. As shown in Fig. 3, PML-WFGALC achieves significantly superior performance compared to other baseline methods. Additionally, PML-WFGALC exhibits significant performance differences across all metrics when compared with LCFS-PML, PML-ND, PLAIN, PML-NI, PAR-MAP, and PML-

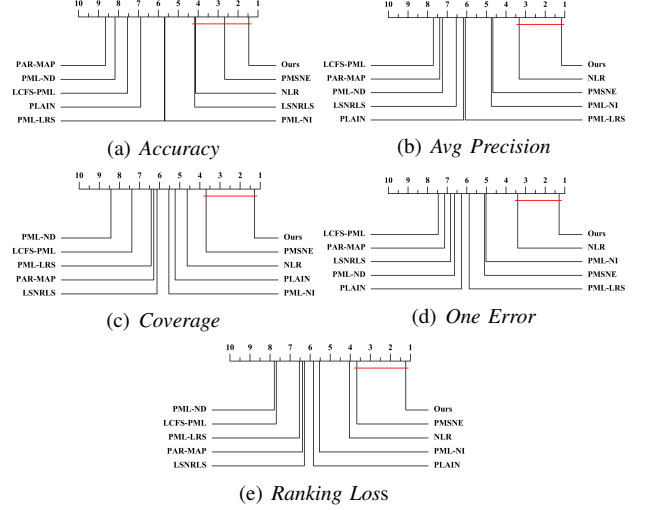


Fig. 3. Comparison of our approach (control algorithm) against nine comparative baselines with the Nemenyi test.

LRS; except for *Accuracy*, PML-WFGALC significantly outperforms LSNRLS across the other four metrics; furthermore, it outperforms NLR significantly in terms of *Coverage* and *Ranking Loss*, and surpasses PMSNE significantly in terms of *Avg Precision* and *One Error*.

C. Ablation Study

An ablation study is useful for examining how various components of the proposed PML-WFGALC method contribute to its prediction performance. Therefore, we compare PML-WFGALC with five variants: v1) the PML-WFGALC model without considering feature graph guided regularization (i.e., $\lambda_2 = 0$); v2) the PML-WFGALC model without considering label correlation guided regularization (i.e., $\lambda_3 = 0$); v3) the PML-WFGALC model with a feature graph constructed by the traditional KNN method; v4) the PML-WFGALC model with a fixed label graph; and v5) the PML-WFGALC model where semantic similarity c_{ij} is computed using ground-truth in the feature graph construction stage. The experimental results on six synthetic datasets for the metrics of *Avg Precision*, *Coverage*, and *Ranking Loss*, are listed in Table VII, where v0 represents the full PML-WFGALC model. From the results of v0, v1, and v2, we can find that both feature-level and label-level graph regularizers improve the LDL model across all datasets and evaluation metrics. When comparing v0 with v3 and v4, v0 achieves better results than these two variants on most datasets—validating the effectiveness of two

TABLE VII
ABLATION STUDY ON SIX SYNTHETIC DATASETS (WHERE THE AVERAGE RESULTS ACROSS THE THREE SYNTHETIC CONFIGURATIONS ARE RECORDED FOR COMPARISON).

DataSets	Metric	v0	v1	v2	v3	v4	v5
Bibtex	Avg Precision	0.5814	0.5733	0.5675	0.5735	0.5675	0.5900
	Coverage	0.1560	0.1824	0.1814	0.1819	0.1814	0.1666
	Ranking	0.0848	0.0955	0.0949	0.0952	0.0949	0.0869
Birds	Avg Precision	0.6253	0.6218	0.6075	0.6211	0.6159	0.6277
	Coverage	0.2649	0.2671	0.2693	0.2669	0.2655	0.2633
	Ranking	0.1669	0.1687	0.1708	0.1690	0.1681	0.1660
Emotions	Avg Precision	0.7701	0.7577	0.7579	0.7597	0.7653	0.7802
	Coverage	0.3258	0.3379	0.3376	0.3362	0.3306	0.3194
	Ranking	0.1984	0.2085	0.2080	0.2067	0.1998	0.1858
Medical	Avg Precision	0.8900	0.8835	0.8835	0.8850	0.8925	0.8965
	Coverage	0.0369	0.0439	0.0439	0.0379	0.0404	0.0369
	Ranking	0.0223	0.0265	0.0265	0.0229	0.0240	0.0218
Stackexchess	Avg Precision	0.4155	0.4039	0.3959	0.4044	0.3979	0.4207
	Coverage	0.2500	0.2728	0.2696	0.2726	0.2704	0.2492
	Ranking	0.1268	0.1413	0.1384	0.1410	0.1384	0.1258
Yeast	Avg Precision	0.7474	0.7444	0.7444	0.7490	0.7461	0.7656
	Coverage	0.4687	0.4957	0.4955	0.4845	0.4924	0.4601
	Ranking	0.1820	0.1915	0.1915	0.1854	0.1897	0.1719

key strategies in our method: weighted feature graphs and adaptive label correlation learning. Finally, a comparison of the results between v0 and v5 reveals that using candidate label sets to calculate semantic similarity c_{ij} has a certain negative impact on the model's performance; this constitutes a limitation of the proposed algorithm. In future work, we will explore more effective ways to compute semantic similarity, aiming to reduce the interference of noisy labels on the model.

D. Denoising Robustness Experiment

To assess the robustness of PML-WFGALC to noisy labels, we set the value of $r\%$ to the range $[25\%, 50\%, \dots, 200\%]$ and conduct experiments on the Birds and Emotions datasets. The aim is to analyze how the noise scale influences the performance of PML-WFGALC. Fig. 4 presents the curves depicting the performance changes of PML-WFGALC as the noise proportion increases. As is evident from the figure, for the Birds dataset: while the overall trend shows that PML-WFGALC's performance fluctuates within a specific range as the noisy label proportion increases (with relatively minor variations), a notable observation is that the model exhibits a slight performance improvement at certain noise proportion levels—even as the noise scale increases. For the Emotions dataset, when the noisy label proportion exceeds 125%, PML-WFGALC's performance undergoes substantial changes; in contrast, when the proportion is below 125%, its performance remains relatively stable. In conclusion, PML-WFGALC demonstrates a certain degree of robustness to noisy labels, with the Birds dataset further reflecting the model's ability to maintain stable performance (and even show minor improvements) under specific noise conditions.

E. Parameter Sensitivity

In this section, we discuss the effects of the three trade-off parameters (i.e., λ_1 , λ_2 , and λ_3), the orthogonality condition regularization parameter μ , the number of nearest neighbors k , and the number of iterations on the proposed method. Since μ should be large enough, we tuned it within the range $\{10^4, 10^5, 10^6, 10^7, 10^8\}$. To facilitate the analysis of

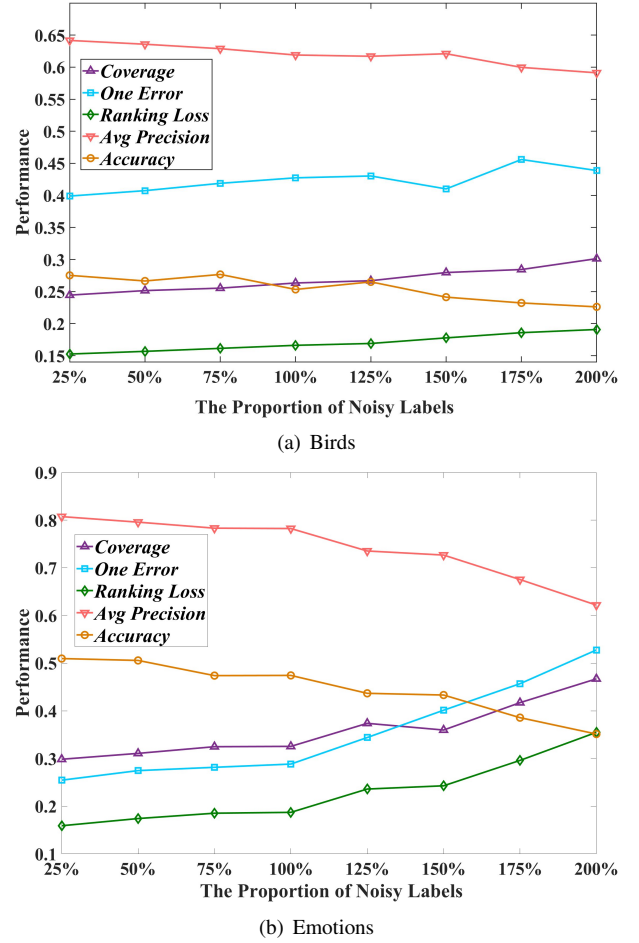


Fig. 4. Robustness analysis of the performance when the proportion of noise in Birds and Emotions datasets is varied.

each parameter, we varied one parameter within its search range while keeping the others fixed. The Avg Precision curves varying with λ_1 , λ_2 , λ_3 , and μ are plotted in Fig. 5¹. From Subfigure 5(a), we can see that as the value of λ_1 increases, the performance of PML-WFGALC first improves and then deteriorates, achieving the best results in the range of $[10^{-1}, 10^1]$. Therefore, we conclude that the model is sensitive to λ_1 . This phenomenon occurs because λ_1 controls the model complexity. A value of λ_1 that is too small will cause the model to overfit, whereas a value that is too large will lead to underfitting. Observing Fig. 5(b) and Fig. 5(c), we can find that the model is not sensitive to λ_2 and λ_3 within a wide range, while the results show an improvement when λ_3 exceeds 10^3 . In Fig. 5(d), when the value of μ exceeds 10^5 , the performance of PML-WFGALC shows a slight decline on most datasets; therefore, μ is set to 10^5 in this study. Fig. 6 shows the performance variation with the number of nearest neighbors k and the number of iterations As can be observed from the figure, the performance of PML-WFGALC shows almost no variation with these two parameters; thus, the model is insensitive to them.

¹Parameter sensitivity experiments conducted on the other four metrics yield similar results

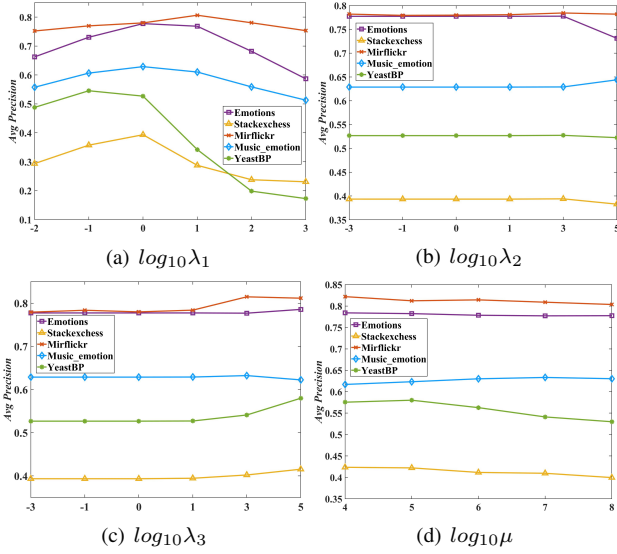


Fig. 5. The Avg Precision results of PML-WFGALC with varying values of λ_1 , λ_2 , λ_3 , and μ on various datasets.

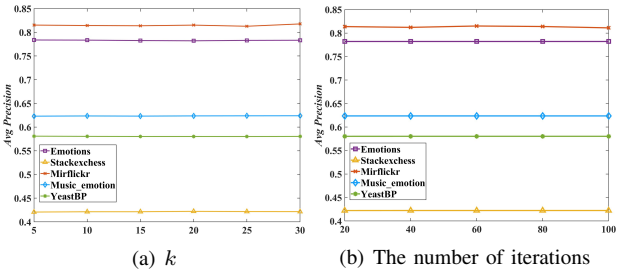


Fig. 6. The Avg Precision results of PML-WFGALC with varying values of k and the number of iterations on various datasets.

F. Convergence Analysis and Time Comparison

To verify the convergence of the proposed model, we plot the objective function values of PML-WFGALC against the number of iterations across six diverse datasets: Bibtex, Birds, Medical, Stackexchess, Music_style, and YeastMF. These convergence curves are presented in Fig. 7, where the Y-axis uses a log-scale to more effectively illustrate the convergence rate of PML-WFGALC. In general, as the number of iterations increases, the loss value decreases, converges quickly, and eventually stabilizes at a satisfactory level. Furthermore, regardless of the dataset size, PML-WFGALC can achieve convergence in fewer than 25 iterations, which confirms the model's strong convergence capability.

Table VIII lists the running time (in seconds) of all comparative methods across datasets. On small datasets (e.g., Birds, Emotions), most algorithms require less training time. For large-sample, low-feature-dimension datasets (e.g., Mirflickr), our method's training speed is slightly slower than that of other methods. However, on datasets with high sample counts and high feature dimensions (e.g., YeastBP), our method is significantly faster. Additionally, PLAIN requires significantly more training time than our method across all datasets. This indicates that compared with deep learning-based approaches (i.e., PLAIN), our method offers greater practicality in real-world applications. Overall, our approach not only achieves

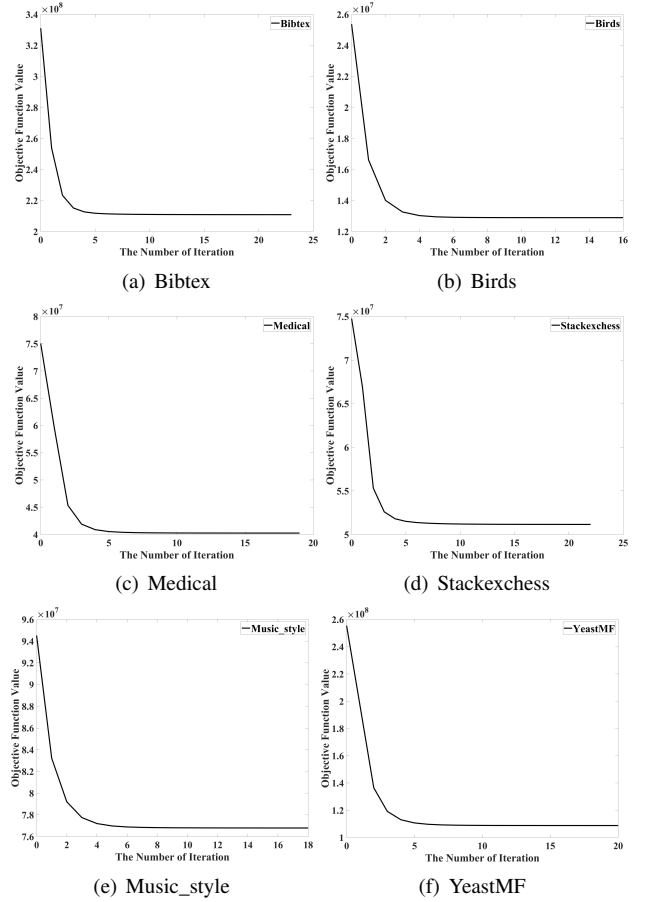


Fig. 7. The convergence curve of PML-WFGALC on six datasets.

superior classification performance but also demonstrates competitive running time compared to existing baselines.

G. Visualization Results

1) *Label Distribution*: To more intuitively demonstrate the effectiveness of our method in noise disambiguation, we show an examples of the label distributions/label confidences learned by PML-WFGALC and several SOTA baselines as in Fig. 8 (another example is shown in Fig. 11 in the appendix). As can be seen from this two figures, all methods achieve higher confidences for relevant labels than for irrelevant ones, this demonstrates their effectiveness in mitigating the impact of noisy labels on multi-label learning. Compared with the other baselines, PML-WFGALC only learns label confidences for candidate labels, while the confidences for non-candidate labels remain 0. This indicates that PML-WFGALC does not generate additional negative information during training and thus can obtain more accurate label distributions.

2) *Feature Graph*: As demonstrated in the ablation study, integrating semantic information into feature graph construction can effectively enhance the prediction performance of PML models. However, using candidate label sets to calculate semantic similarity c_{ij} makes the model susceptible to noisy labels, which in turn degrades its performance. To demonstrate how noisy labels impact the model during the feature graph construction stage, we plot heatmaps of the adjacency matrices

TABLE VIII
RUNNING TIME(IN SECONDS) OF DIFFERENT METHODS.

DataSets	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	22.872	119.479	2818.6	14.176	16.558	40.606	240.922	46.533	1743.956	13.097
Birds	0.067	0.062	95.845	0.106	0.048	0.065	12.115	0.349	47.418	0.036
Emotions	0.184	0.076	16.488	0.023	0.003	0.110	13.313	0.148	0.893	0.055
Medical	1.899	1.493	465.973	3.038	4.321	5.291	22.375	4.806	1.500	1.350
Stackexchess	1.578	4.627	97.466	1.549	2.640	18.447	109.748	17.281	35.328	1.545
Yeast	5.391	15.030	155.549	0.130	0.024	0.908	47.462	1.149	1.471	0.972
Mirflickr	2.867	62.552	15.234	0.218	0.032	1.123	48.282	1.208	1.928	5.898
Music_emotion	5.885	135.175	533.544	0.344	1.073	0.412	67.239	2.500	122.948	5.977
Music_style	5.845	134.390	664.542	0.344	0.059	0.238	65.388	2.267	211.221	6.196
YeastBP	18.549	4158.500	2.34E+04	97.266	186.010	46.355	153.083	253.181	2149.827	21.375
YeastCC	5.049	646.912	2.51E+04	70.847	100.604	17.982	40.920	107.681	142.672	15.560
YeastMF	6.377	581.693	1.31E+04	70.706	91.566	16.834	42.050	93.194	118.691	15.733

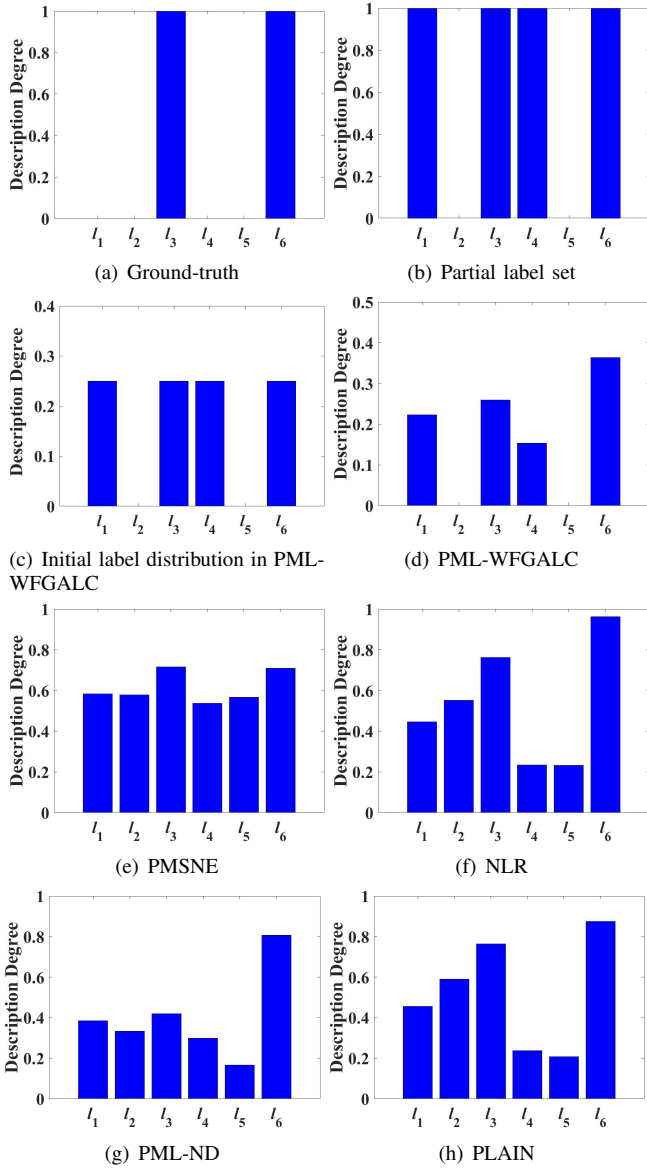


Fig. 8. The first example of label distributions or label confidences learned by PML-WFGALC and other methods on the Emotions dataset.

for feature graphs constructed with/without semantic similarity and under different noise densities. The experimental results are presented in Fig. 9, where white regions indicate the k nearest neighbors of the corresponding nodes. By comparing Subfigures 9(a) and 9(c) with 9(b), we can observe that when the number of noisy labels is small (i.e., $r = 50$), the feature graph constructed using candidate labels is more similar to the one built with ground truth than the feature graph generated via unsupervised KNN. This finding indicates that although noisy labels introduce some impact, the advantages of integrating candidate label information into feature graph construction far outweigh its drawbacks. Fig. 9(d) shows that as the number of noisy labels increases, the discrepancy between feature graphs constructed using candidate label sets and ground truth becomes more pronounced, which in turn exerts a greater negative impact on model performance. This observation indicates that semantic similarity c_{ij} is highly sensitive to noise density. In future work, we will explore more effective approaches to compute c_{ij} , aiming to reduce its sensitivity to noise and enhance the algorithm's robustness.

3) *Label Correlation*: To demonstrate the effectiveness of adaptive label correlation learning, we plot heatmaps of the initial label correlation matrix, the learned label correlation matrix, and the label correlation matrix computed using ground truth, as shown in Fig. 10. As can be observed from the figure, although adaptive label correlation learning fails to match the ground-truth label correlation matrix, it effectively mitigates the interference caused by noisy labels. For instance, the true correlation coefficient between class 1 and class 3 is 0.1450; however, due to the influence of noisy labels, their correlation coefficient computed using the candidate label set increases to 0.1981. After learning via our adaptive method, this correlation coefficient is reduced to 0.1735. These results demonstrate that adaptive learning enables us to obtain label correlations closer to those derived from ground truth, thereby validating the effectiveness of the proposed algorithm in this study.

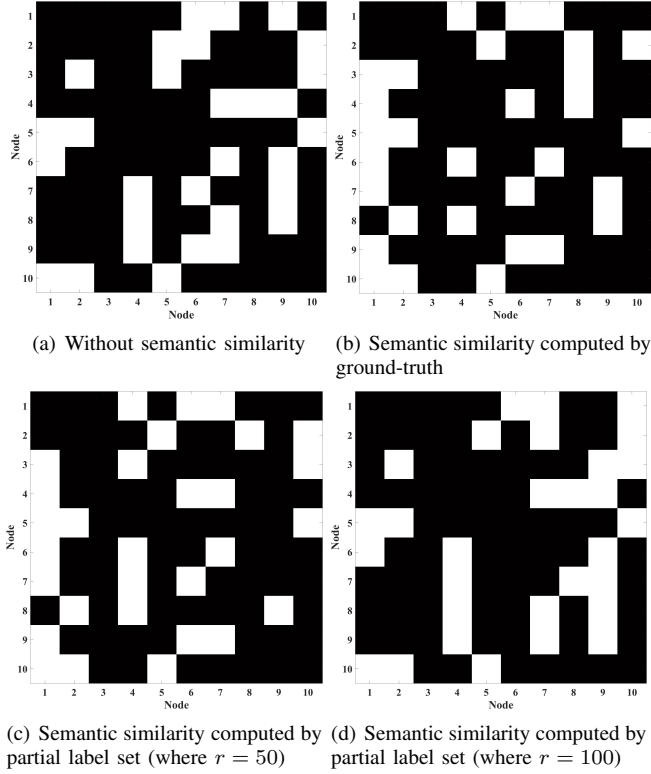


Fig. 9. An example of the constructed feature graphs with/without semantic information and under different noise densities, where the number of nearest neighbors is set as 3.

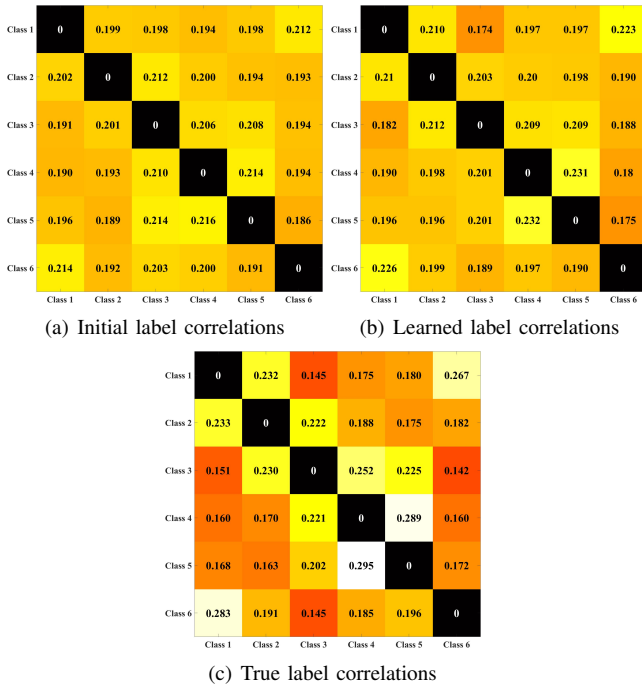


Fig. 10. An example of the learned label correlations.

V. CONCLUSION AND DISCUSSION

In this paper, we first elucidate the significance of semantic information for constructing the feature space's topological structure and further uncover how effectively bootstrapping label correlations benefits the prediction model. Building upon these insights, we have proposed a novel disambiguation method for partial multi-label learning, which is guided by a weighted feature graph and adaptive label correlation learning. The proposed method makes three key contributions: (1) It constructs a nonlinear predictive function that is regularized by a feature-label dual graph. This enables the learning of the latent label distribution, facilitating the identification of ground-truth labels with a high label probability; (2) It induces a weighted k-nearest neighbors method grounded in semantic information. This approach helps in constructing a more accurate topological structure of the feature space; (3) It establishes an adaptive label correlation learning mechanism to derive more precise label relationships. We formulated an objective function based on these principles and devised an efficient iterative optimization algorithm to solve it. Our comprehensive evaluation across 18 synthetic and 6 real-world PML datasets demonstrates consistent superiority over state-of-the-art methods.

However, PML-WFGALC has two notable limitations. First, its semantic similarity calculation relies solely on candidate labels, rendering it highly sensitive to noisy label counts and severely degrading feature graph quality. Second, it fails to leverage label-specific features: different labels often correlate with distinct discriminative features (e.g., edge features for 'has_wing' vs. texture features for 'has_beak' in Birds), and the shared feature space cannot exploit this heterogeneity. Future work will address these limitations: (1) Explore dynamic semantic similarity learning by fusing candidate labels with sample feature similarity (e.g., deep embeddings) to reduce noise reliance; (2) Develop a label-specific feature selection module (via attention or sparse learning) to build an adaptive feature space. These efforts aim to further enhance PML-WFGALC's robustness and practicality.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no conflict of interest.

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APPENDIX

The experimental results of all comparative methods on *Accuracy* and *One Error* are listed in Table IX and Table X, respectively. And the other example of the learned label distributions or label confidences is presented in Fig. 11.

TABLE IX
EXPERIMENTAL RESULTS COMPARING ALL APPROACHES IN TERMS OF *Accuracy* (WITH BEST PERFORMANCE INDICATED IN BOLDFACE).

DataSets	Noise Ratio	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	50%	.2311±.0029	.2296±.0054	.1261±.0015	.1811±.0034	.0587±.0019	.2299±.0056	.0873±.0020	.2029±.0055	.0964±.0042	.2223±.0114
	100%	.2079±.0052	.2215±.0081	.1223±.0015	.1701±.0069	.0535±.0011	.2153±.0102	.0984±.0024	.1876±.0069	.0916±.0049	.2066±.0112
	150%	.1863±.0069	.2097±.0062	.1167±.0026	.1638±.0071	.0477±.0013	.1899±.0068	.1036±.0056	.1727±.0077	.0923±.0029	.1795±.0065
Birds	50%	.2665±.0184	.2485±.0302	.2090±.0277	.2374±.0170	.1585±.0087	.2513±.0270	.1682±.0224	.2630±.0143	.1516±.0193	.2528±.0417
	100%	.2533±.0370	.2185±.0394	.2036±.0166	.2295±.0328	.1340±.0095	.2031±.0333	.1613±.0169	.2300±.0283	.1691±.0247	.2158±.0160
	150%	.2413±.0204	.2298±.0254	.2008±.0327	.2177±.0274	.1259±.0124	.1916±.0172	.1665±.0105	.1956±.0183	.1978±.0163	.1978±.0168
Emotions	50%	.5065±.0389	.4785±.0244	.4493±.0411	.4701±.0364	.4634±.0365	.5027±.0255	.5130±.0421	.4642±.0366	.3629±.0259	.4547±.0322
	100%	.4768±.0367	.4567±.0217	.4410±.0351	.4474±.0415	.4629±.0370	.4548±.0222	.4102±.0241	.4230±.0271	.3702±.0243	.4275±.0203
	150%	.4406±.0243	.4367±.0106	.3900±.0356	.4216±.0259	.3637±.0409	.4035±.0167	.3271±.0170	.3343±.0222	.3280±.0367	.3495±.0199
Medical	50%	.2719±.0078	.2647±.0122	.1925±.0181	.3055±.0247	.2066±.0148	.2214±.0238	.1903±.0039	.2580±.0082	.2014±.0116	.2064±.0273
	100%	.2608±.0142	.2537±.0237	.1832±.0244	.3066±.0196	.1800±.0199	.1849±.0381	.1996±.0130	.2486±.0135	.2059±.0204	.2109±.0396
	150%	.2510±.0079	.2214±.0034	.1718±.0123	.2868±.0208	.1679±.0160	.1751±.0315	.2045±.0151	.2269±.0078	.2051±.0328	.2167±.0295
Stackexchess	50%	.0549±.0165	.0497±.0178	.0314±.0059	.0457±.0078	.0385±.0093	.0576±.0166	.0105±.0018	.0366±.0101	.0067±.0009	.0372±.0057
	100%	.0510±.0147	.0455±.0145	.0310±.0059	.0422±.0071	.0343±.0098	.0535±.0125	.0106±.0020	.0338±.0084	.0109±.0040	.0367±.0072
	150%	.0486±.0140	.0427±.0131	.0309±.0054	.0414±.0074	.0267±.0049	.0542±.0148	.0103±.0018	.0317±.0083	.0120±.0038	.0373±.0090
Yeast	50%	.3460±.0035	.3449±.0093	.2863±.0070	.3356±.0075	.2944±.0052	.3392±.0052	.3351±.0066	.2540±.0052	.1941±.0064	.2678±.0121
	100%	.3417±.0066	.3391±.0035	.2756±.0044	.3285±.0054	.2893±.0022	.3362±.0042	.3312±.0051	.2763±.0048	.2305±.0125	.2677±.0070
	150%	.3300±.0042	.3288±.0061	.2697±.0055	.3194±.0042	.2808±.0055	.3246±.0025	.3113±.0045	.3085±.0066	.2322±.0110	.2351±.0084
Mirflickr	/	.4885±.0109	.4570±.0132	.3419±.0072	.4151±.0066	.3345±.0073	.3961±.0215	.4425±.0100	.4154±.0204	.3673±.0161	.3568±.0049
Music_emotion	/	.3273±.0040	.3131±.0074	.2723±.0062	.3038±.0041	.1882±.0068	.3244±.0067	.3074±.0060	.2504±.0112	.1161±.0140	.2818±.0076
Music_style	/	.2907±.0086	.2693±.0054	.2299±.0067	.2656±.0053	.0509±.0034	.2865±.0077	.2403±.0018	.2664±.0052	.0981±.0084	.2345±.0075
YeastBP	/	.1910±.0603	.1853±.0571	.0871±.0241	.1597±.0488	.0450±.0118	.1206±.0239	.1464±.0401	.1396±.0421	.0968±.0162	.1709±.0445
YeastCC	/	.2778±.0623	.2675±.0555	.1208±.0341	.2286±.0322	.0965±.0193	.1750±.0190	.2316±.0602	.1986±.0456	.1661±.0284	.2047±.0504
YeastMF	/	.2052±.0477	.2010±.0427	.0885±.0164	.1724±.0347	.0791±.0155	.1388±.0200	.1529±.0459	.1429±.0346	.1002±.0150	.1839±.0338
avg_rank		1.46	2.67	7.54	4.13	8.17	4.17	6.88	5.67	8.65	5.69

TABLE X
EXPERIMENTAL RESULTS COMPARING ALL APPROACHES IN TERMS OF *One Error* (WITH BEST PERFORMANCE INDICATED IN BOLDFACE).

DataSets	Noise Ratio	Ours	PMSNE	LCFS-PML	NLR	PML-ND	LSNRLS	PLAIN	PML-NI	PAR-MAP	PML-LRS
Bibtex	50%	.3535±.0053	.4026±.0038	.3973±.0067	.3629±.0059	.4069±.0194	.3663±.0078	.5244±.0113	.3757±.0080	.4702±.0031	.3689±.0053
	100%	.3546±.0063	.4020±.0068	.3997±.0060	.3666±.0036	.4065±.0215	.3777±.0090	.5175±.0109	.3851±.0088	.4645±.0088	.3770±.0065
	150%	.3705±.0032	.4084±.0060	.4116±.0049	.3774±.0055	.3985±.0178	.3888±.0049	.5064±.0147	.4019±.0063	.4776±.0079	.4076±.0056
Birds	50%	.4074±.0612	.4274±.0785	.4475±.0749	.3988±.0601	.4530±.0177	.4529±.0575	.4473±.0473	.4215±.0646	.5187±.0500	.4443±.0580
	100%	.4274±.0631	.4616±.0500	.4418±.0770	.4303±.0557	.4900±.0377	.5158±.0655	.4531±.0547	.4472±.0442	.5357±.0361	.4644±.0589
	150%	.4101±.0362	.4474±.0583	.4587±.0610	.4273±.0640	.4588±.0501	.5129±.0357	.4701±.0320	.4331±.0454	.4815±.0526	.4728±.0809
Emotions	50%	.2664±.0364	.3778±.0218	.3491±.0607	.3053±.0445	.3102±.0216	.3490±.0306	.3068±.0511	.3187±.0345	.3117±.0701	.3423±.0455
	100%	.2867±.0505	.3271±.0321	.3171±.0240	.3187±.0393	.3306±.0198	.3206±.0360	.3187±.0220	.3459±.0419	.3507±.0326	.3238±.0446
	150%	.3728±.0336	.4384±.0915	.4416±.0690	.4400±.0798	.4064±.0325	.4720±.0530	.3734±.0347	.4739±.0643	.4230±.0492	.4249±.0719
Medical	50%	.1339±.0124	.1400±.0195	.2894±.0232	.1390±.0161	.1922±.0094	.1943±.0212	.2004±.0172	.1400±.0179	.2260±.0182	.1852±.0390
	100%	.1380±.0164	.1533±.0261	.2975±.0410	.1350±.0126	.1923±.0110	.2710±.0266	.1973±.0293	.1441±.0224	.2240±.0273	.1903±.0607
	150%	.1401±.0103	.1625±.0199	.3549±.0363	.1421±.0171	.2055±.0138	.2820±.0663	.1994±.0261	.1585±.0174	.2270±.0273	.1850±.0273
Stackexchess	50%	.4886±.0394	.5963±.0605	.5335±.0316	.4910±.0213	.4704±.0187	.6095±.0768	.7458±.0439	.5352±.0350	.7236±.0314	.5532±.0294
	100%	.4862±.0383	.5927±.0627	.5341±.0277	.5012±.0250	.4967±.0070	.5690±.0700	.7458±.0435	.5388±.0247	.7075±.0369	.5514±.0219
	150%	.4952±.0274	.5909±.0549	.5401±.0225	.5060±.0105	.5266±.0219	.6012±.0468	.7452±.0443	.5406±.0190	.7015±.0366	.5389±.0107
Yeast	50%	.2193±.0180	.2404±.0135	.4237±.0203	.3339±.0159	.3533±.0227	.2478±.0123	.2644±.0262	.3225±.0154	.2226±.0095	.3951±.0167
	100%	.2325±.0168	.2396±.0154	.4845±.0275	.3463±.0162	.3819±.0179	.2474±.0174	.3020±.0180	.2528±.0150	.2441±.0089	.4108±.0102
	150%	.2400±.0246	.2482±.0210	.5813±.0284	.4034±.0311	.4141±.0294	.2611±.0138	.3169±.0274	.3173±.0179	.2743±.0182	.4568±.0123
Mirflickr	/	.1808±.0040	.1850±.0131	.4942±.0149	.3634±.0148	.3358±.0051	.2458±.0364	.2150±.0129	.2800±.0042	.1540±.0118	.2874±.0145
Music_emotion	/	.4256±.0105	.5295±.0145	.5094±.0114	.4649±.0194	.5580±.0528	.5686±.0163	.4193±.0243	.4762±.0193	.5920±.0215	.4958±.0152
Music_style	/	.3446±.0060	.3958±.0049	.3666±.0105	.3458±.0060	.3971±.0042	.3815±.0082	.3518±.0071	.3457±.0080	.4119±.0086	.3559±.0081
YeastBP	/	.5275±.1047	.5324±.1218	.7021±.0877	.5508±.1157	.6619±.0963	.6243±.0928	.6213±.1106	.6130±.0975	.5498±.1276	.5982±.1240
YeastCC	/	.4475±.1143	.4504±.1157	.6866±.0416	.4684±.1124	.5523±.0934	.5350±.0995	.5190±.0826	.5301±.0887	.4886±.0832	.5141±.1049
YeastMF	/	.6543±.0724	.6315±.0611	.7895±.0321	.6458±.0546	.7244±.0445	.7221±.0365	.6581±.0632	.7032±.0394	.7213±.0409	.6762±.0569
avg_rank		1.29	5.10	7.46	3.40	6.63	6.83	6.27	5.02	7.13	5.88

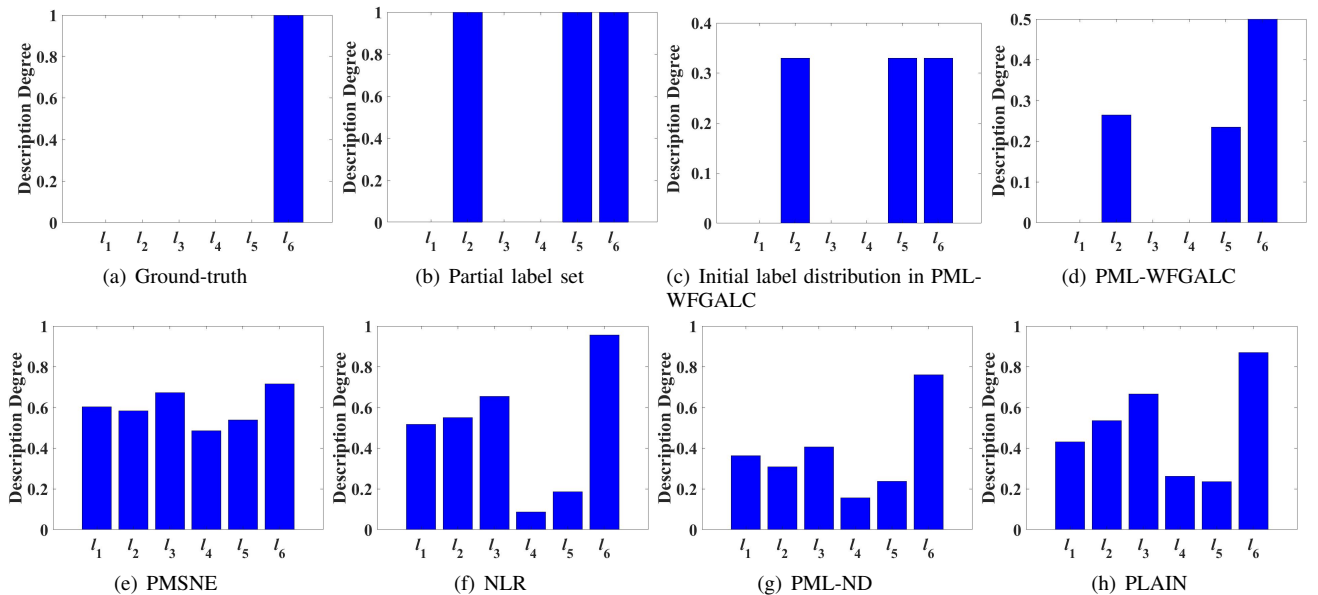


Fig. 11. The other example of label distributions or label confidences learned by PML-WFGALC and other methods on the Emotions dataset.